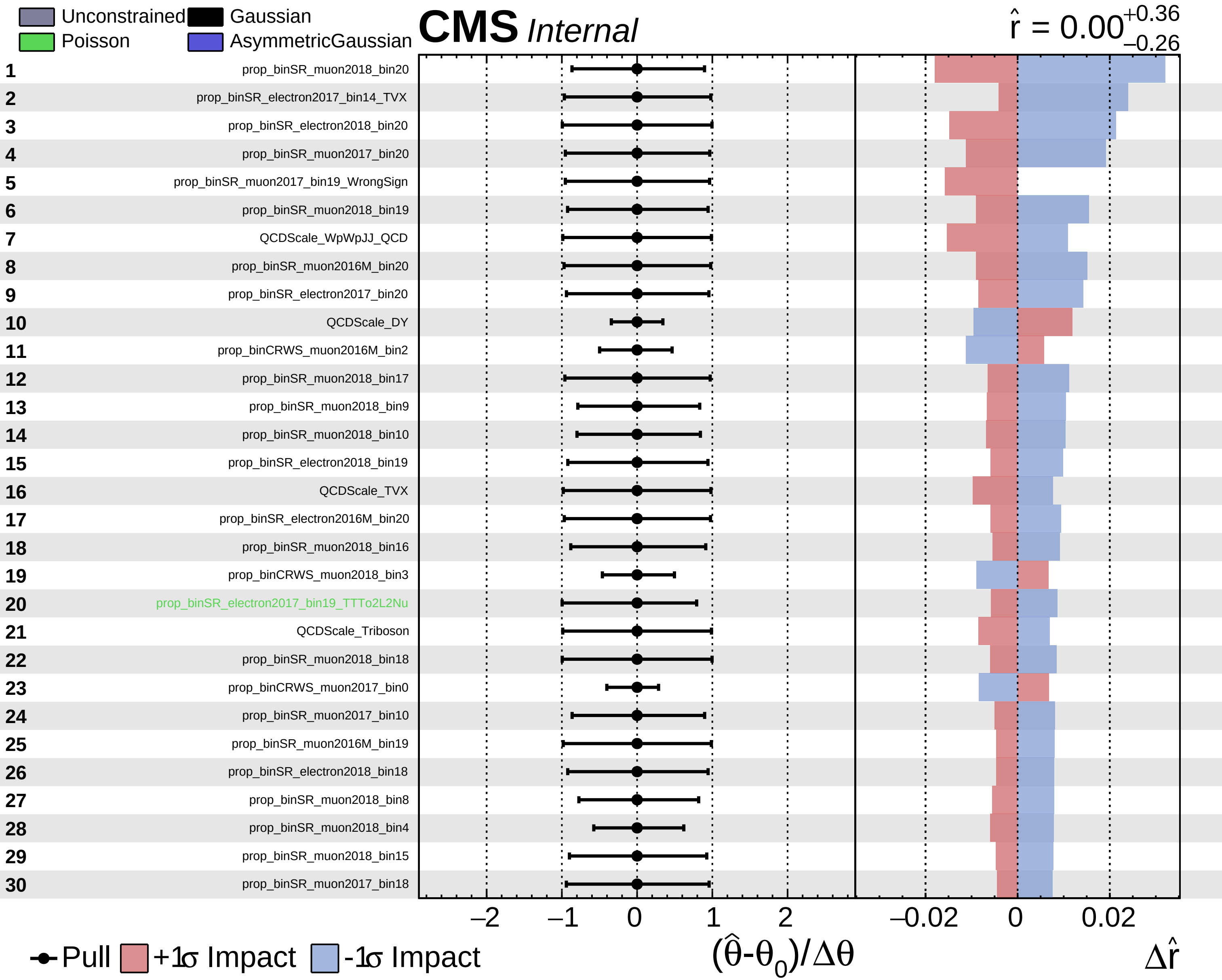
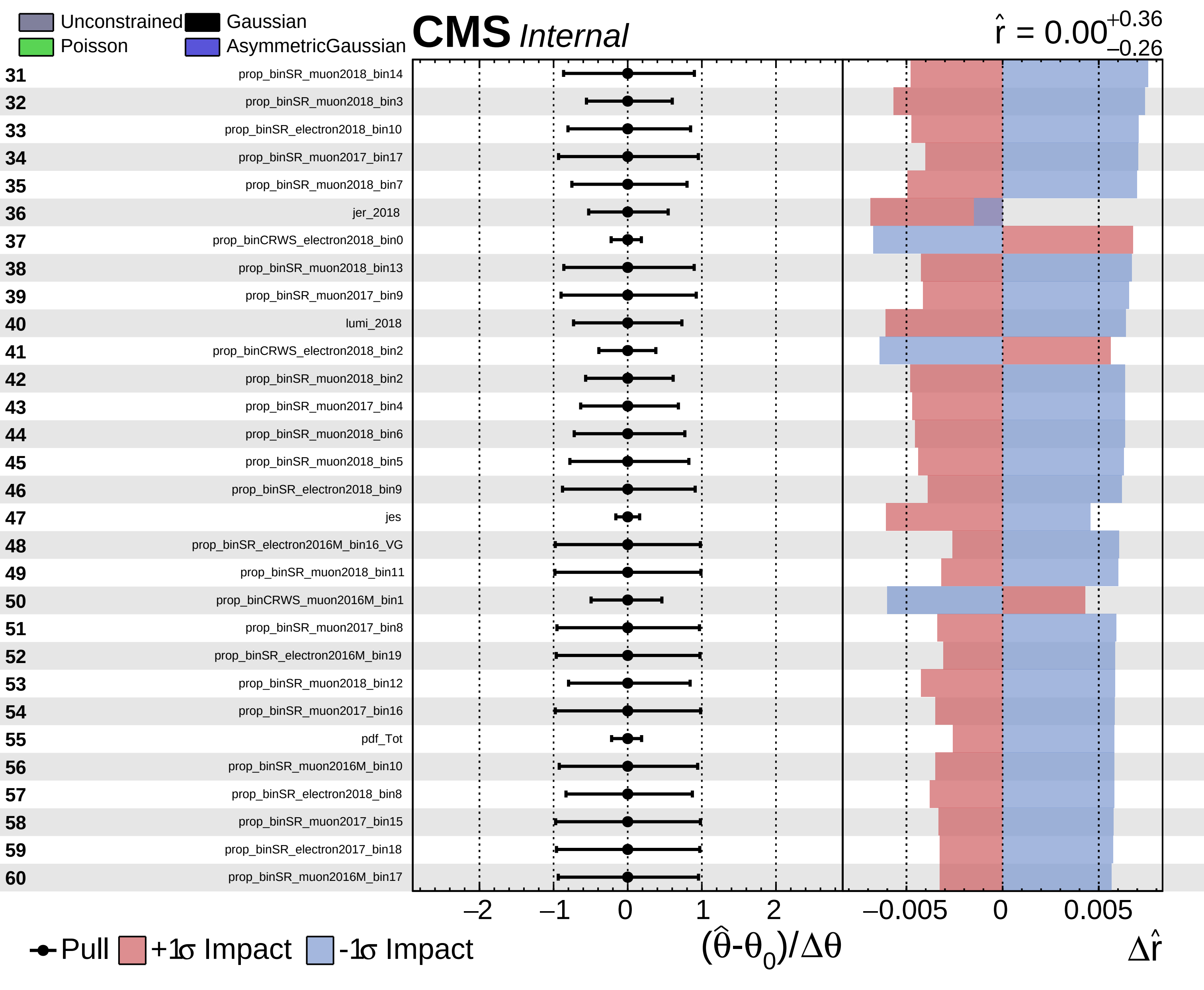
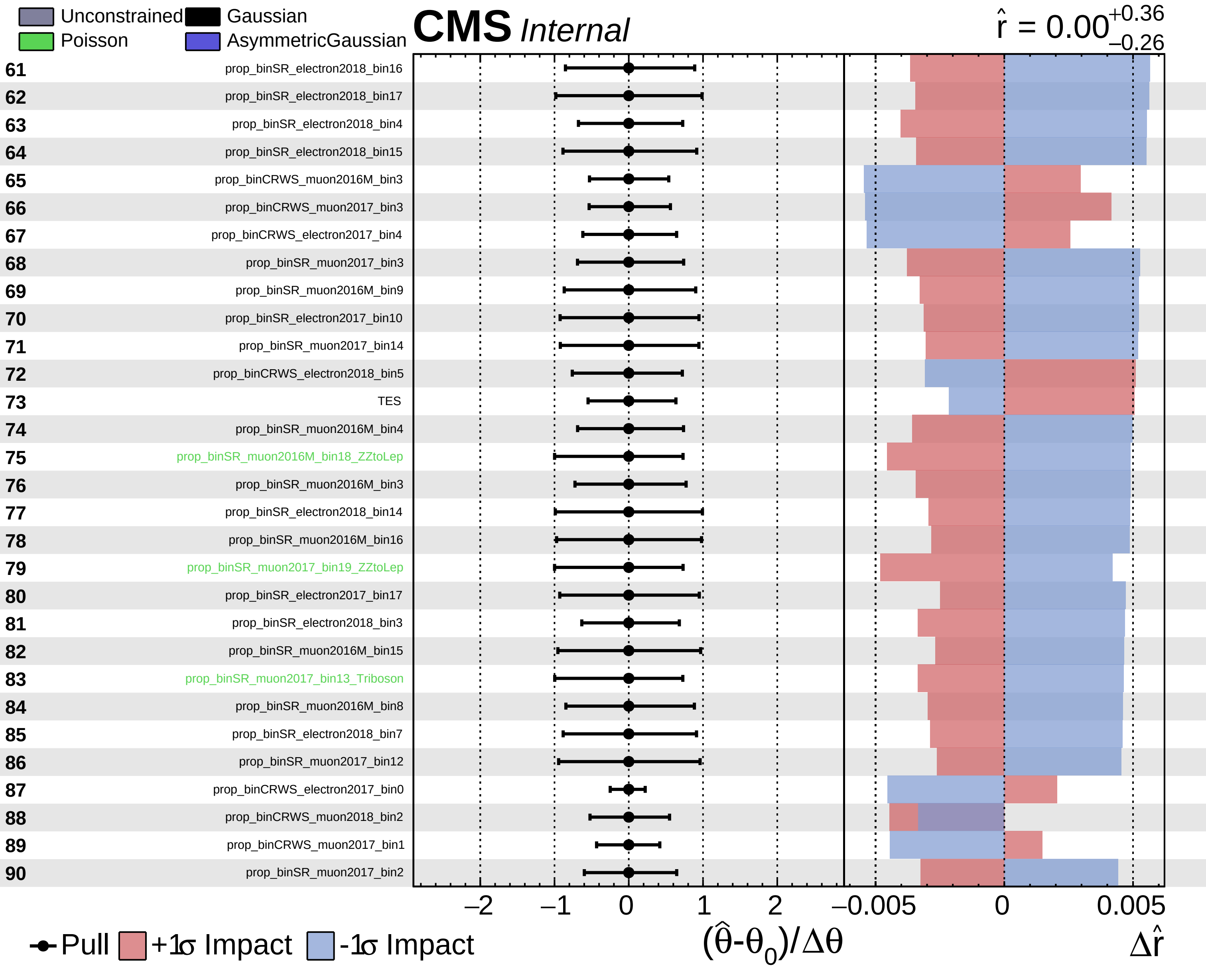


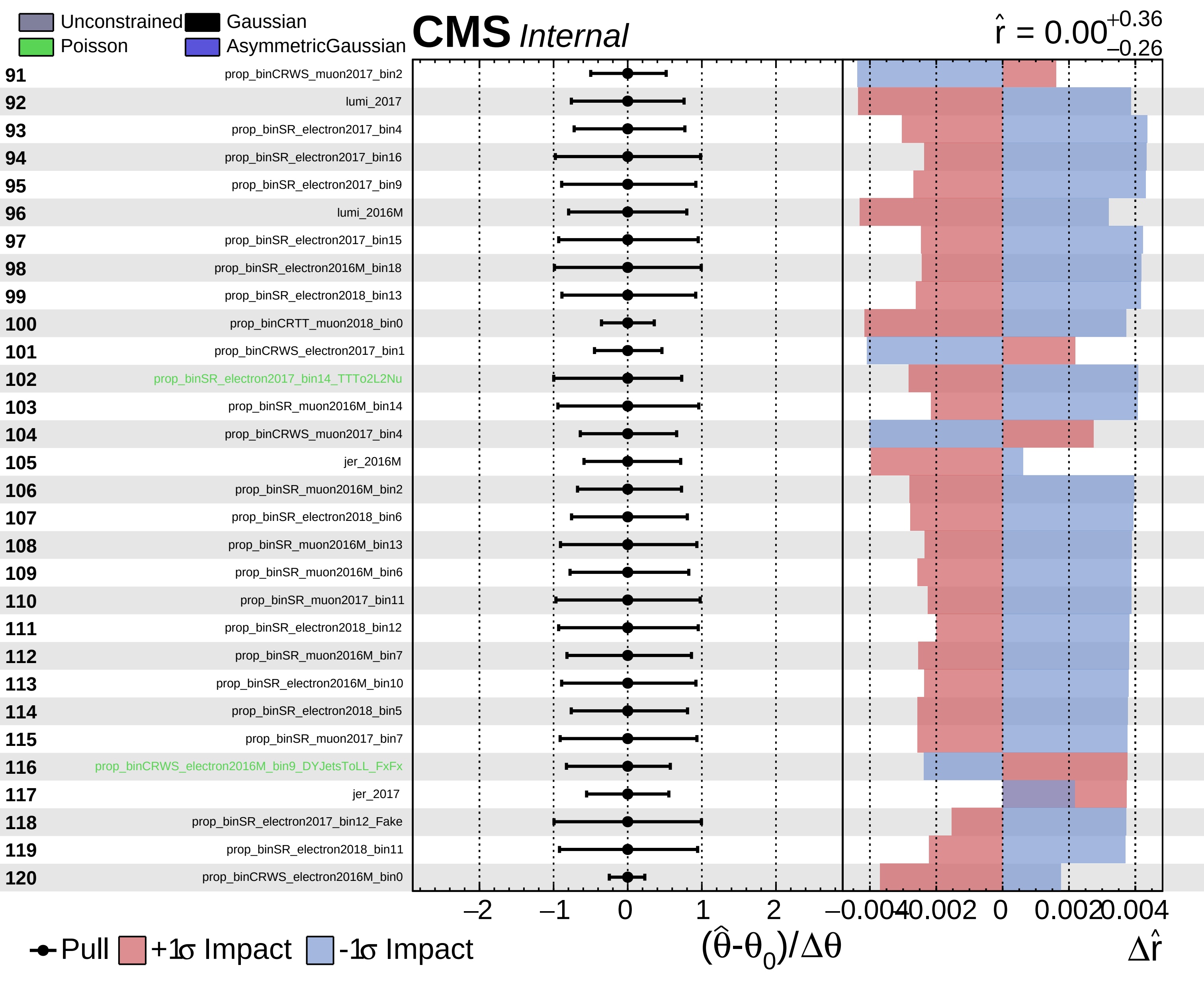
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.36}_{-0.26}$
1 prop_binSR_muon2018_bin20
2 prop_binSR_electron2017_bin14_TVX
3 prop_binSR_electron2018_bin20
4 prop_binSR_muon2017_bin20
5 prop_binSR_muon2017_bin19_WrongSign
6 prop_binSR_muon2018_bin19
7 QCDScale_WpWpJJ_QCD
8 prop_binSR_muon2016M_bin20
9 prop_binSR_electron2017_bin20
10 QCDScale_DY
11 prop_binCRWS_muon2016M_bin2
12 prop_binSR_muon2018_bin17
13 prop_binSR_muon2018_bin9
14 prop_binSR_muon2018_bin10
15 prop_binSR_electron2018_bin19
16 QCDScale_TVX
17 prop_binSR_electron2016M_bin20
18 prop_binSR_muon2018_bin16
19 prop_binCRWS_muon2018_bin3
20 prop_binSR_electron2017_bin19_TTTo2L2Nu
21 QCDScale_Triboson
22 prop_binSR_muon2018_bin18
23 prop_binCRWS_muon2017_bin0
24 prop_binSR_muon2017_bin10
25 prop_binSR_muon2016M_bin19
26 prop_binSR_electron2018_bin18
27 prop_binSR_muon2018_bin8
28 prop_binSR_muon2018_bin4
29 prop_binSR_muon2018_bin15
30 prop_binSR_muon2017_bin18
−2 −1 0 1 2
−0.02 0 0.02
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



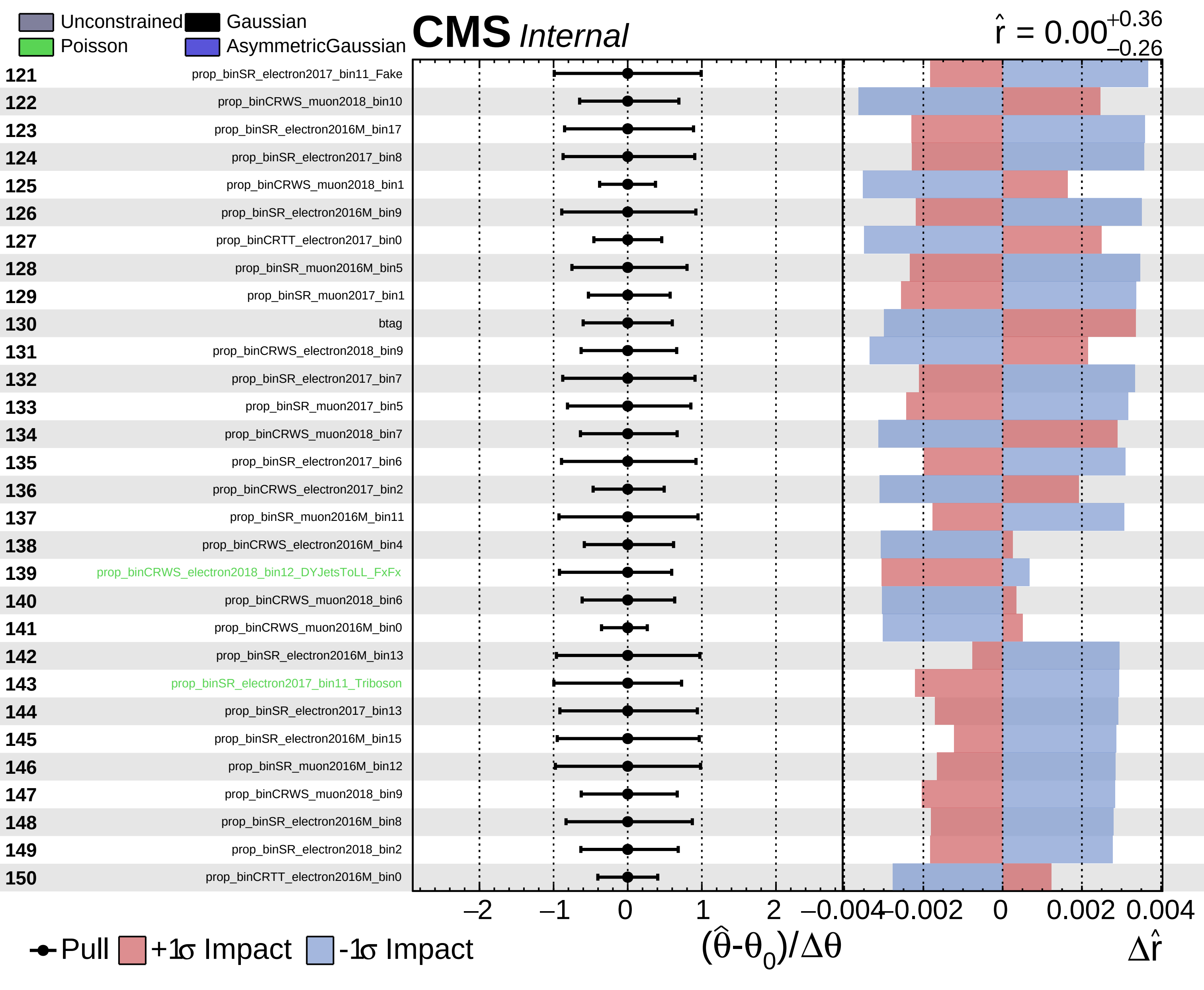
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.36}_{-0.26}$
31 prop_binSR_muon2018_bin14
32 prop_binSR_muon2018_bin3
33 prop_binSR_electron2018_bin10
34 prop_binSR_muon2017_bin17
35 prop_binSR_muon2018_bin7
36 jer_2018
37 prop_binCRWS_electron2018_bin0
38 prop_binSR_muon2018_bin13
39 prop_binSR_muon2017_bin9
40 lumi_2018
41 prop_binCRWS_electron2018_bin2
42 prop_binSR_muon2018_bin2
43 prop_binSR_muon2017_bin4
44 prop_binSR_muon2018_bin6
45 prop_binSR_muon2018_bin5
46 prop_binSR_electron2018_bin9
47 jes
48 prop_binSR_electron2016M_bin16_VG
49 prop_binSR_muon2018_bin11
50 prop_binCRWS_muon2016M_bin1
51 prop_binSR_muon2017_bin8
52 prop_binSR_electron2016M_bin19
53 prop_binSR_muon2018_bin12
54 prop_binSR_muon2017_bin16
55 pdf_Tot
56 prop_binSR_muon2016M_bin10
57 prop_binSR_electron2018_bin8
58 prop_binSR_muon2017_bin15
59 prop_binSR_electron2017_bin18
60 prop_binSR_muon2016M_bin17
−2 −1 0 1 2
−0.005 0 0.005
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



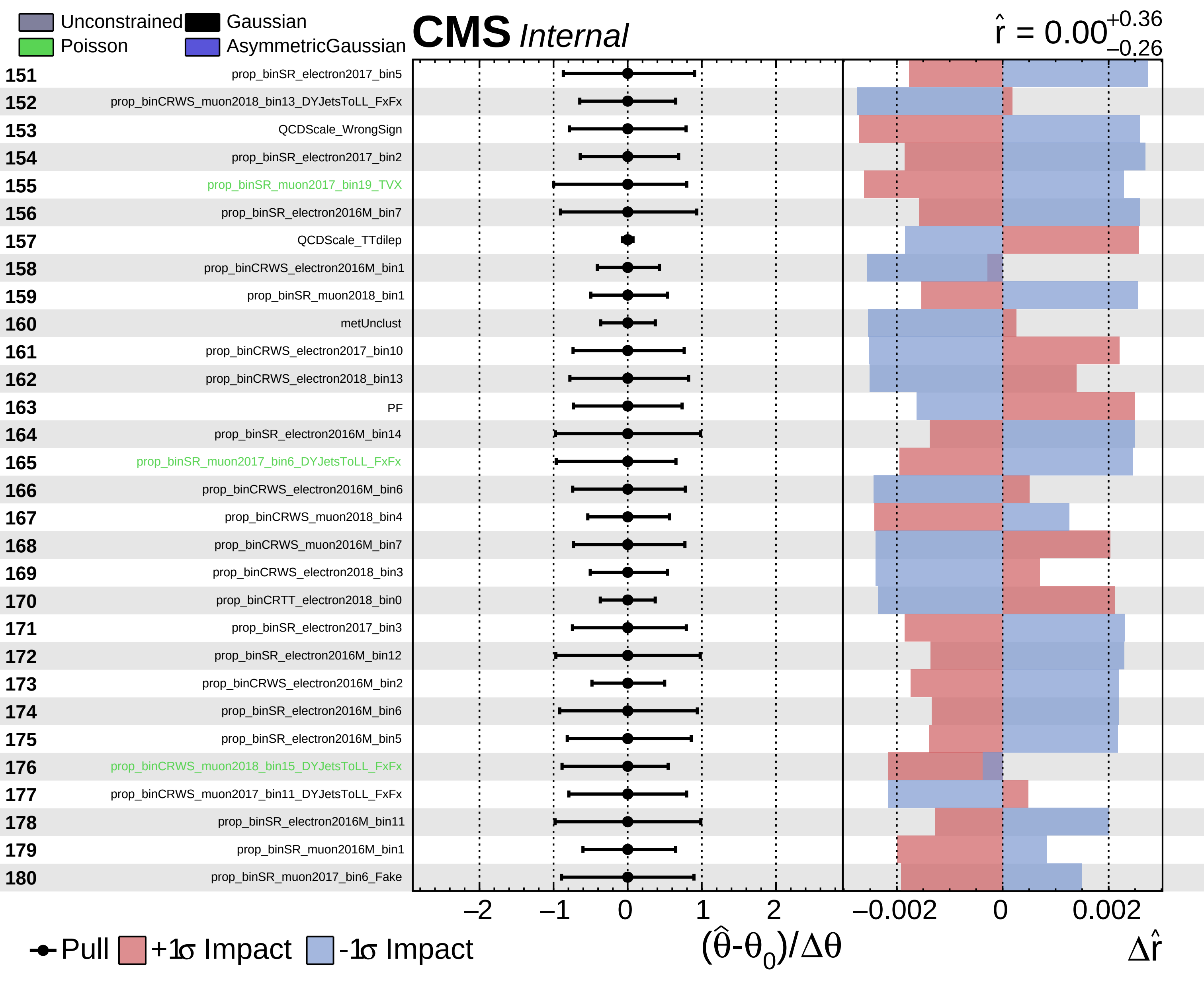
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.36}_{-0.26}$
61 prop_binSR_electron2018_bin16
62 prop_binSR_electron2018_bin17
63 prop_binSR_electron2018_bin4
64 prop_binSR_electron2018_bin15
65 prop_binCRWS_muon2016M_bin3
66 prop_binCRWS_muon2017_bin3
67 prop_binCRWS_electron2017_bin4
68 prop_binSR_muon2017_bin3
69 prop_binSR_muon2016M_bin9
70 prop_binSR_electron2017_bin10
71 prop_binSR_muon2017_bin14
72 prop_binCRWS_electron2018_bin5
73 TES
74 prop_binSR_muon2016M_bin4
75 prop_binSR_muon2016M_bin18_ZZtoLep
76 prop_binSR_muon2016M_bin3
77 prop_binSR_electron2018_bin14
78 prop_binSR_muon2016M_bin16
79 prop_binSR_muon2017_bin19_ZZtoLep
80 prop_binSR_electron2017_bin17
81 prop_binSR_electron2018_bin3
82 prop_binSR_muon2016M_bin15
83 prop_binSR_muon2017_bin13_Triboson
84 prop_binSR_muon2016M_bin8
85 prop_binSR_electron2018_bin7
86 prop_binSR_muon2017_bin12
87 prop_binCRWS_electron2017_bin0
88 prop_binCRWS_muon2018_bin2
89 prop_binCRWS_muon2017_bin1
90 prop_binSR_muon2017_bin2
−2
−1
0
1
2
−0.005
0
0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



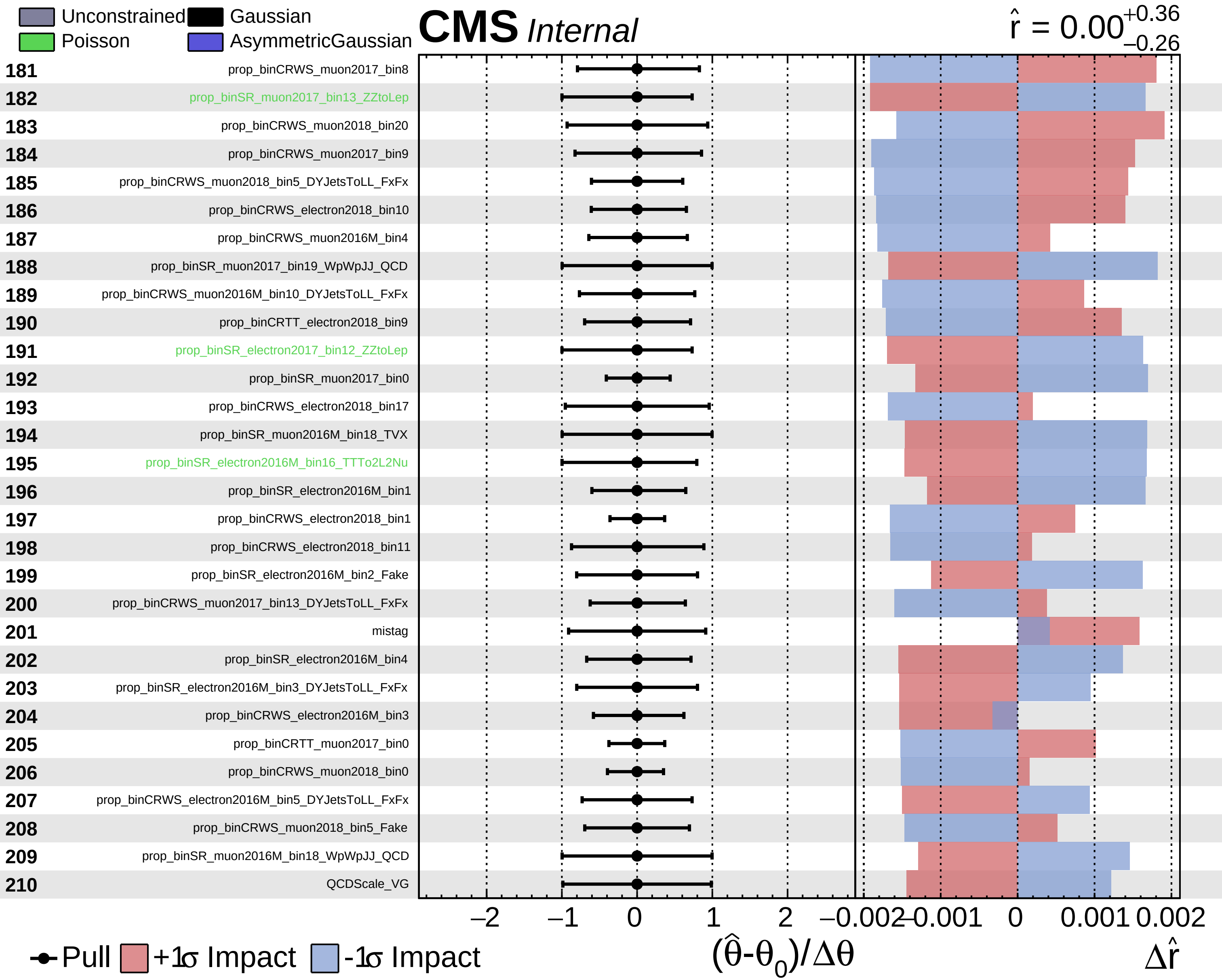
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.36}_{-0.26}$
91 prop_binCRWS_muon2017_bin2
92 lumi_2017
93 prop_binSR_electron2017_bin4
94 prop_binSR_electron2017_bin16
95 prop_binSR_electron2017_bin9
96 lumi_2016M
97 prop_binSR_electron2017_bin15
98 prop_binSR_electron2016M_bin18
99 prop_binSR_electron2018_bin13
100 prop_binCRTT_muon2018_bin0
101 prop_binCRWS_electron2017_bin1
102 prop_binSR_electron2017_bin14_TTTo2L2Nu
103 prop_binSR_muon2016M_bin14
104 prop_binCRWS_muon2017_bin4
105 jer_2016M
106 prop_binSR_muon2016M_bin2
107 prop_binSR_electron2018_bin6
108 prop_binSR_muon2016M_bin13
109 prop_binSR_muon2016M_bin6
110 prop_binSR_muon2017_bin11
111 prop_binSR_electron2018_bin12
112 prop_binSR_muon2016M_bin7
113 prop_binSR_electron2016M_bin10
114 prop_binSR_electron2018_bin5
115 prop_binSR_muon2017_bin7
116 prop_binCRWS_electron2016M_bin9_DYJetsToLL_FxFx
117 jer_2017
118 prop_binSR_electron2017_bin12_Fake
119 prop_binSR_electron2018_bin11
120 prop_binCRWS_electron2016M_bin0
−2
−1
0
1
2
−0.004
−0.002
0
0.002
0.004
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



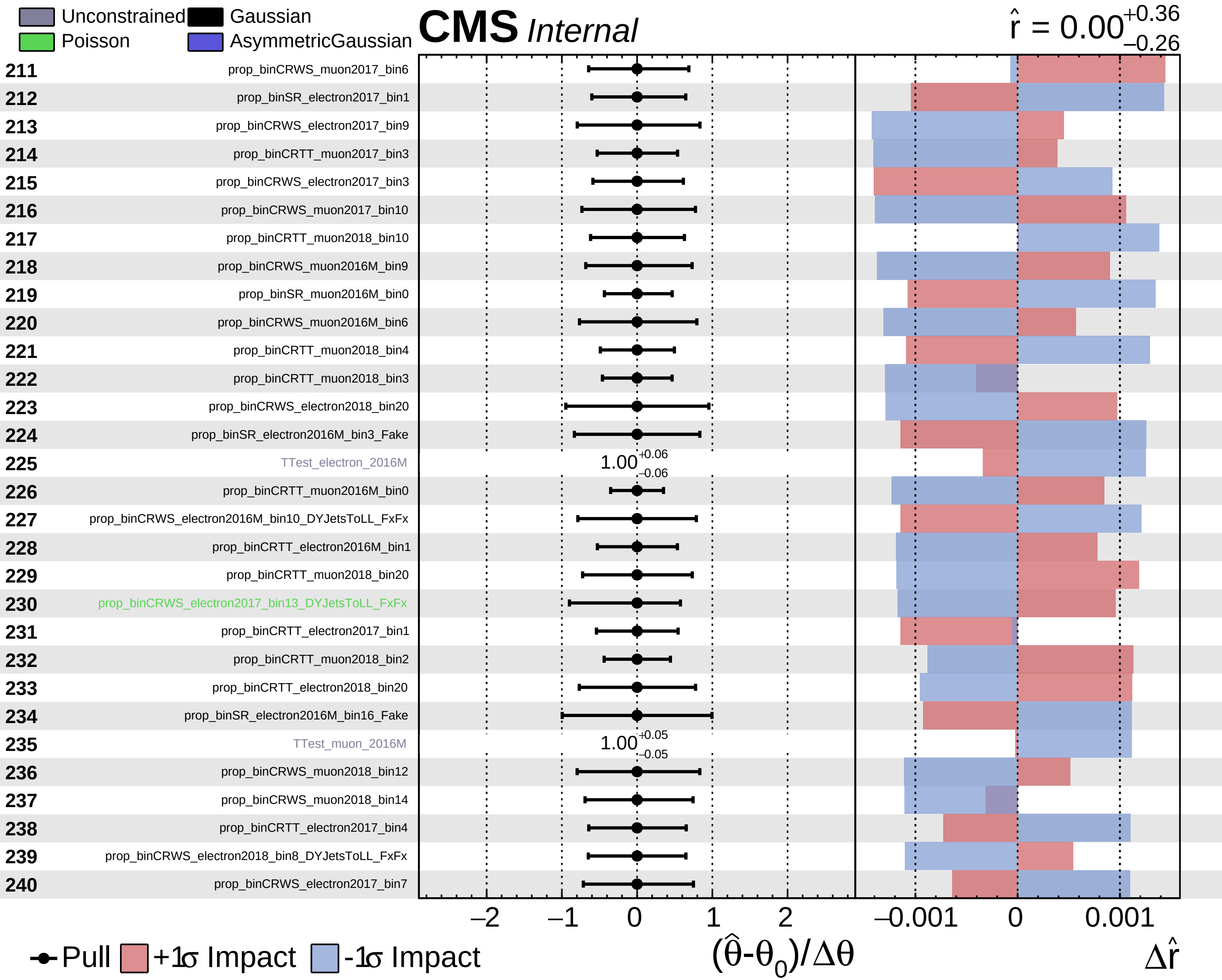
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.36}_{-0.26}$
121 prop_binSR_electron2017_bin11_Fake
122 prop_binCRWS_muon2018_bin10
123 prop_binSR_electron2016M_bin17
124 prop_binSR_electron2017_bin8
125 prop_binCRWS_muon2018_bin1
126 prop_binSR_electron2016M_bin9
127 prop_binCRTT_electron2017_bin0
128 prop_binSR_muon2016M_bin5
129 prop_binSR_muon2017_bin1
130 btag
131 prop_binCRWS_electron2018_bin9
132 prop_binSR_electron2017_bin7
133 prop_binSR_muon2017_bin5
134 prop_binCRWS_muon2018_bin7
135 prop_binSR_electron2017_bin6
136 prop_binCRWS_electron2017_bin2
137 prop_binSR_muon2016M_bin11
138 prop_binCRWS_electron2016M_bin4
139 prop_binCRWS_electron2018_bin12_DYJetsToLL_FxFx
140 prop_binCRWS_muon2018_bin6
141 prop_binCRWS_muon2016M_bin0
142 prop_binSR_electron2016M_bin13
143 prop_binSR_electron2017_bin11_Triboson
144 prop_binSR_electron2017_bin13
145 prop_binSR_electron2016M_bin15
146 prop_binSR_muon2016M_bin12
147 prop_binCRWS_muon2018_bin9
148 prop_binSR_electron2016M_bin8
149 prop_binSR_electron2018_bin2
150 prop_binCRTT_electron2016M_bin0
−2 −1 0 1 2
−0.004 −0.002 0 0.002 0.004
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



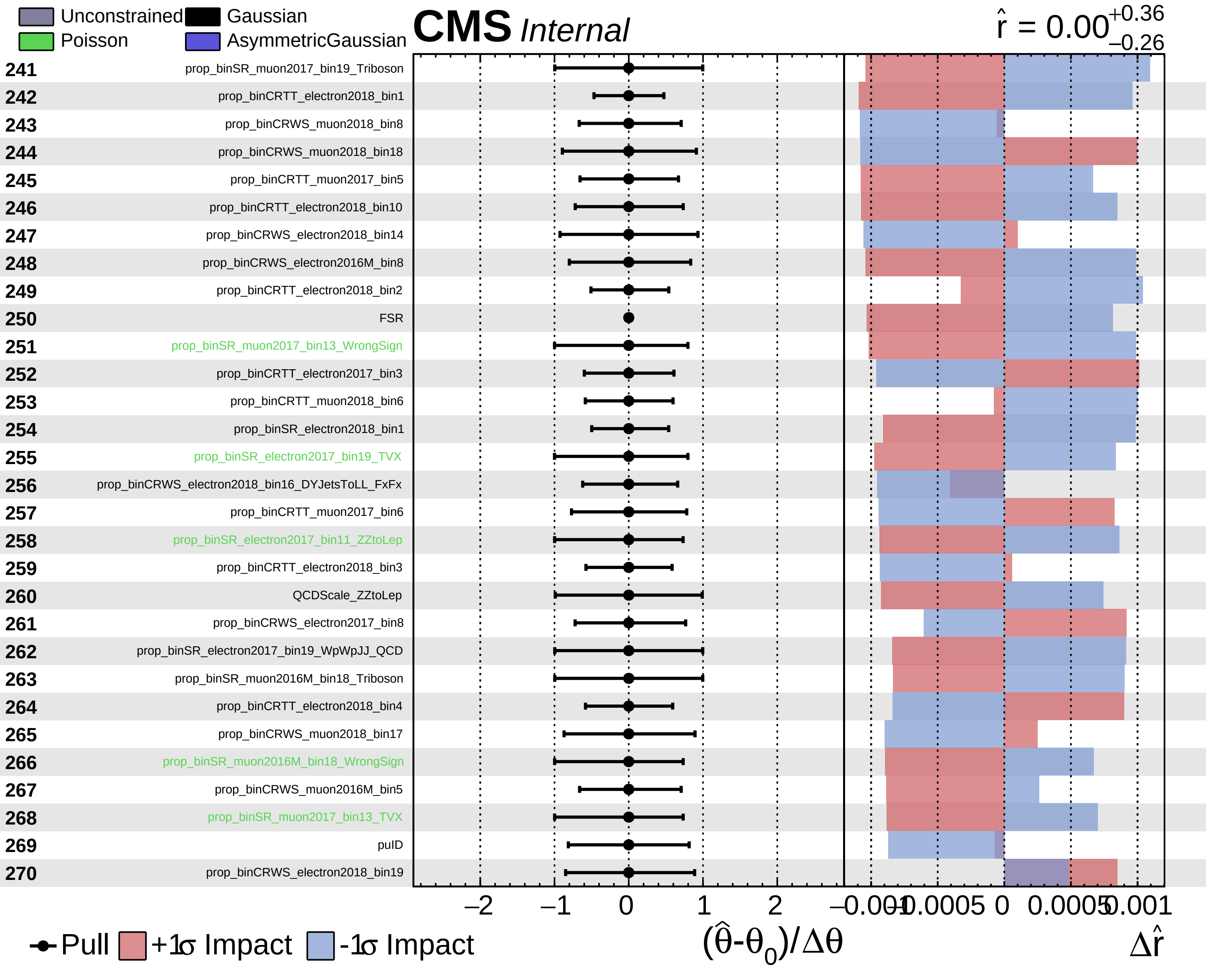
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.36}_{-0.26}$
151 prop_binSR_electron2017_bin5
152 prop_binCRWS_muon2018_bin13_DYJetsToLL_FxFx
153 QCDScale_WrongSign
154 prop_binSR_electron2017_bin2
155 prop_binSR_muon2017_bin19_TVX
156 prop_binSR_electron2016M_bin7
157 QCDScale_TTdilep
158 prop_binCRWS_electron2016M_bin1
159 prop_binSR_muon2018_bin1
160 metUnclust
161 prop_binCRWS_electron2017_bin10
162 prop_binCRWS_electron2018_bin13
163 PF
164 prop_binSR_electron2016M_bin14
165 prop_binSR_muon2017_bin6_DYJetsToLL_FxFx
166 prop_binCRWS_electron2016M_bin6
167 prop_binCRWS_muon2018_bin4
168 prop_binCRWS_muon2016M_bin7
169 prop_binCRWS_electron2018_bin3
170 prop_binCRTT_electron2018_bin0
171 prop_binSR_electron2017_bin3
172 prop_binSR_electron2016M_bin12
173 prop_binCRWS_electron2016M_bin2
174 prop_binSR_electron2016M_bin6
175 prop_binSR_electron2016M_bin5
176 prop_binCRWS_muon2018_bin15_DYJetsToLL_FxFx
177 prop_binCRWS_muon2017_bin11_DYJetsToLL_FxFx
178 prop_binSR_electron2016M_bin11
179 prop_binSR_muon2016M_bin1
180 prop_binSR_muon2017_bin6_Fake
−2 −1 0 1 2
−0.002 0 0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



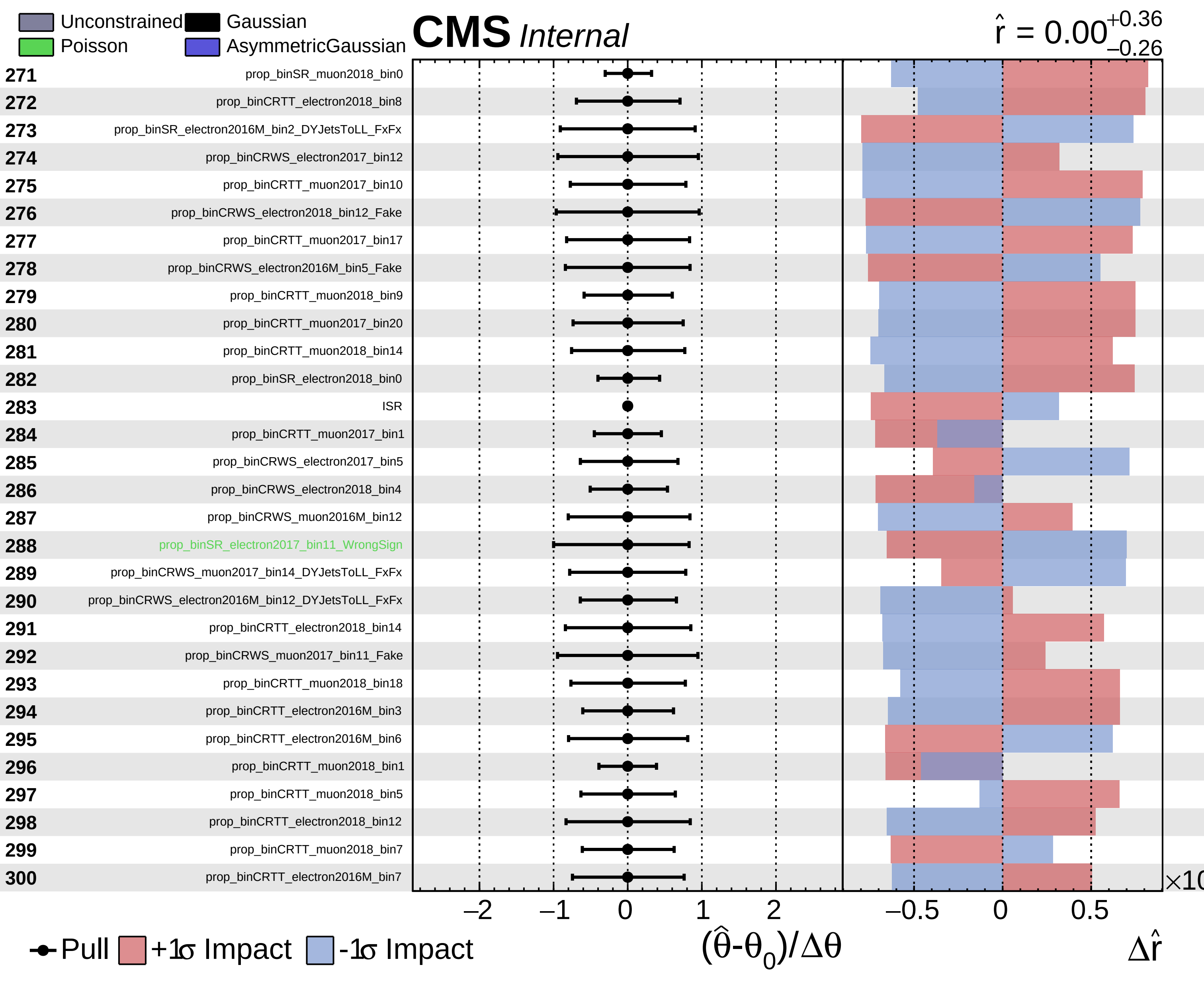
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.36}_{-0.26}$
181 prop_binCRWS_muon2017_bin8
182 prop_binSR_muon2017_bin13_ZZtoLep
183 prop_binCRWS_muon2018_bin20
184 prop_binCRWS_muon2017_bin9
185 prop_binCRWS_muon2018_bin5_DYJetsToLL_FxFx
186 prop_binCRWS_electron2018_bin10
187 prop_binCRWS_muon2016M_bin4
188 prop_binSR_muon2017_bin19_WpWpJJ_QCD
189 prop_binCRWS_muon2016M_bin10_DYJetsToLL_FxFx
190 prop_binCRTT_electron2018_bin9
191 prop_binSR_electron2017_bin12_ZZtoLep
192 prop_binSR_muon2017_bin0
193 prop_binCRWS_electron2018_bin17
194 prop_binSR_muon2016M_bin18_TVX
195 prop_binSR_electron2016M_bin16_TTTo2L2Nu
196 prop_binSR_electron2016M_bin1
197 prop_binCRWS_electron2018_bin1
198 prop_binCRWS_electron2018_bin11
199 prop_binSR_electron2016M_bin2_Fake
200 prop_binCRWS_muon2017_bin13_DYJetsToLL_FxFx
201 mistag
202 prop_binSR_electron2016M_bin4
203 prop_binSR_electron2016M_bin3_DYJetsToLL_FxFx
204 prop_binCRWS_electron2016M_bin3
205 prop_binCRTT_muon2017_bin0
206 prop_binCRWS_muon2018_bin0
207 prop_binCRWS_electron2016M_bin5_DYJetsToLL_FxFx
208 prop_binCRWS_muon2018_bin5_Fake
209 prop_binSR_muon2016M_bin18_WpWpJJ_QCD
210 QCDScale_VG
−2 −1 0 1 2
−0.002 −0.001 0 0.001 0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



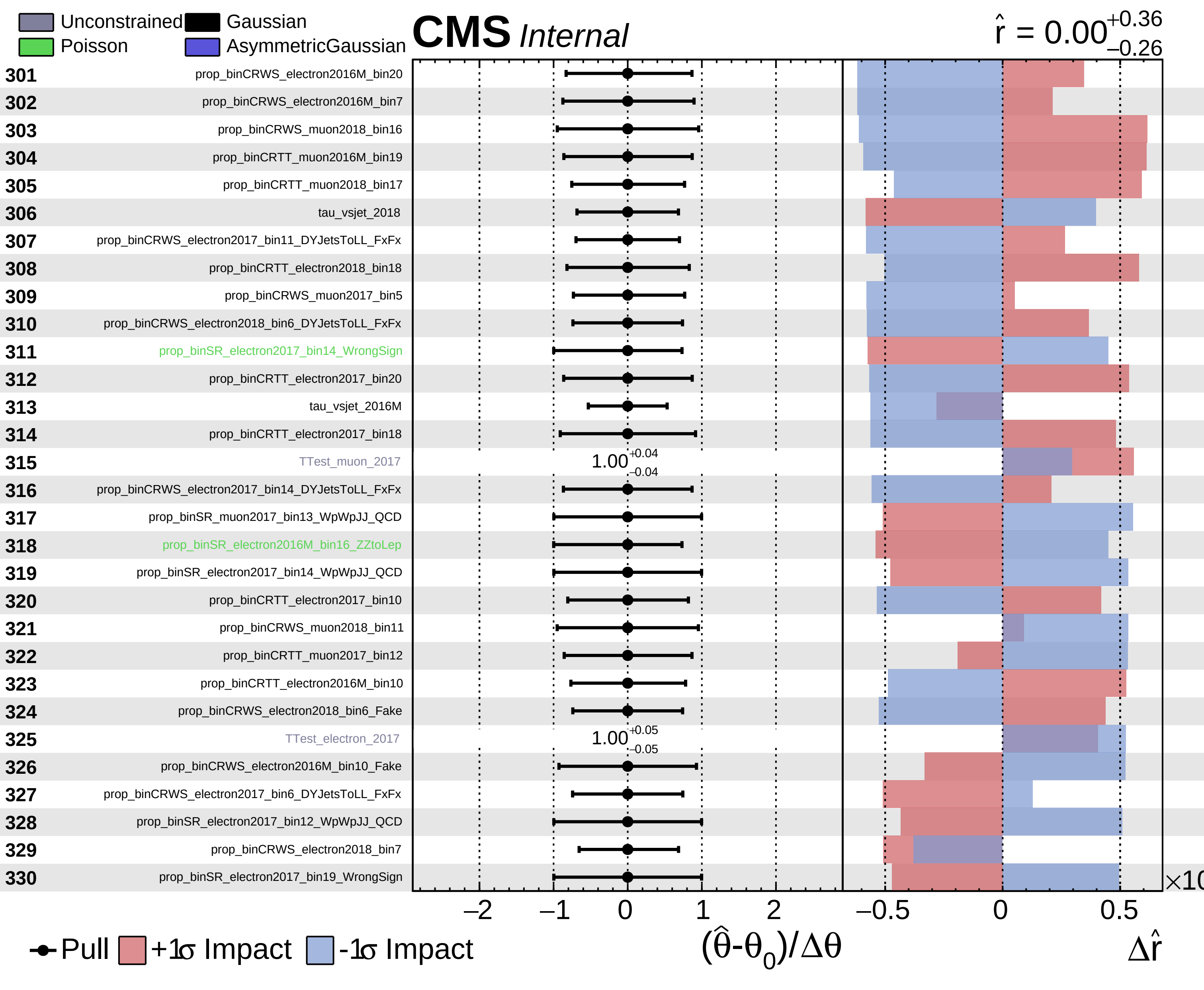
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.36 −0.26
211 prop_binCRWS_muon2017_bin6
212 prop_binSR_electron2017_bin1
213 prop_binCRWS_electron2017_bin9
214 prop_binCRTT_muon2017_bin3
215 prop_binCRWS_electron2017_bin3
216 prop_binCRWS_muon2017_bin10
217 prop_binCRTT_muon2018_bin10
218 prop_binCRWS_muon2016M_bin9
219 prop_binSR_muon2016M_bin0
220 prop_binCRWS_muon2016M_bin6
221 prop_binCRTT_muon2018_bin4
222 prop_binCRTT_muon2018_bin3
223 prop_binCRWS_electron2018_bin20
224 prop_binSR_electron2016M_bin3_Fake
225 TTest_electron_2016M 1.00 +0.06 −0.06
226 prop_binCRTT_muon2016M_bin0
227 prop_binCRWS_electron2016M_bin10_DYJetsToLL_FxFx
228 prop_binCRTT_electron2016M_bin1
229 prop_binCRTT_muon2018_bin20
230 prop_binCRWS_electron2017_bin13_DYJetsToLL_FxFx
231 prop_binCRTT_electron2017_bin1
232 prop_binCRTT_muon2018_bin2
233 prop_binCRTT_electron2018_bin20
234 prop_binSR_electron2016M_bin16_Fake
235 TTest_muon_2016M 1.00 +0.05 −0.05
236 prop_binCRWS_muon2018_bin12
237 prop_binCRWS_muon2018_bin14
238 prop_binCRTT_electron2017_bin4
239 prop_binCRWS_electron2018_bin8_DYJetsToLL_FxFx
240 prop_binCRWS_electron2017_bin7
−2 −1 0 1 2
−0.001 0 0.001
Pull +1σ Impact -1σ Impact
(θ̂-θ0)/Δθ
Δr̂



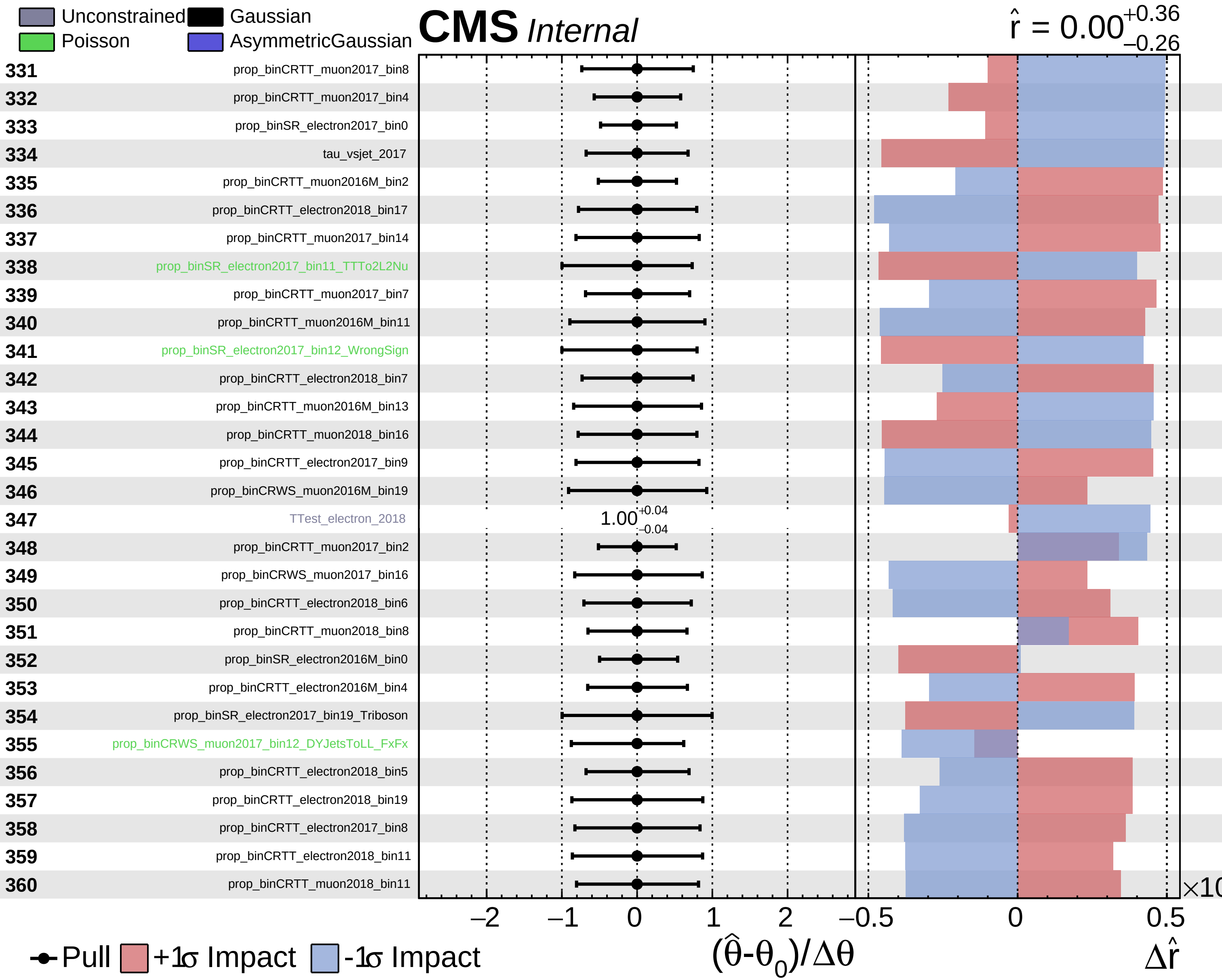
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.36}_{-0.26}$
241 prop_binSR_muon2017_bin19_Triboson
242 prop_binCRTT_electron2018_bin1
243 prop_binCRWS_muon2018_bin8
244 prop_binCRWS_muon2018_bin18
245 prop_binCRTT_muon2017_bin5
246 prop_binCRTT_electron2018_bin10
247 prop_binCRWS_electron2018_bin14
248 prop_binCRWS_electron2016M_bin8
249 prop_binCRTT_electron2018_bin2
250 FSR
251 prop_binSR_muon2017_bin13_WrongSign
252 prop_binCRTT_electron2017_bin3
253 prop_binCRTT_muon2018_bin6
254 prop_binSR_electron2018_bin1
255 prop_binSR_electron2017_bin19_TVX
256 prop_binCRWS_electron2018_bin16_DYJetsToLL_FxFx
257 prop_binCRTT_muon2017_bin6
258 prop_binSR_electron2017_bin11_ZZtoLep
259 prop_binCRTT_electron2018_bin3
260 QCDScale_ZZtoLep
261 prop_binCRWS_electron2017_bin8
262 prop_binSR_electron2017_bin19_WpWpJJ_QCD
263 prop_binSR_muon2016M_bin18_Triboson
264 prop_binCRTT_electron2018_bin4
265 prop_binCRWS_muon2018_bin17
266 prop_binSR_muon2016M_bin18_WrongSign
267 prop_binCRWS_muon2016M_bin5
268 prop_binSR_muon2017_bin13_TVX
269 puID
270 prop_binCRWS_electron2018_bin19
−2
−1
0
1
2
−0.001
−0.0005
0
0.0005
0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



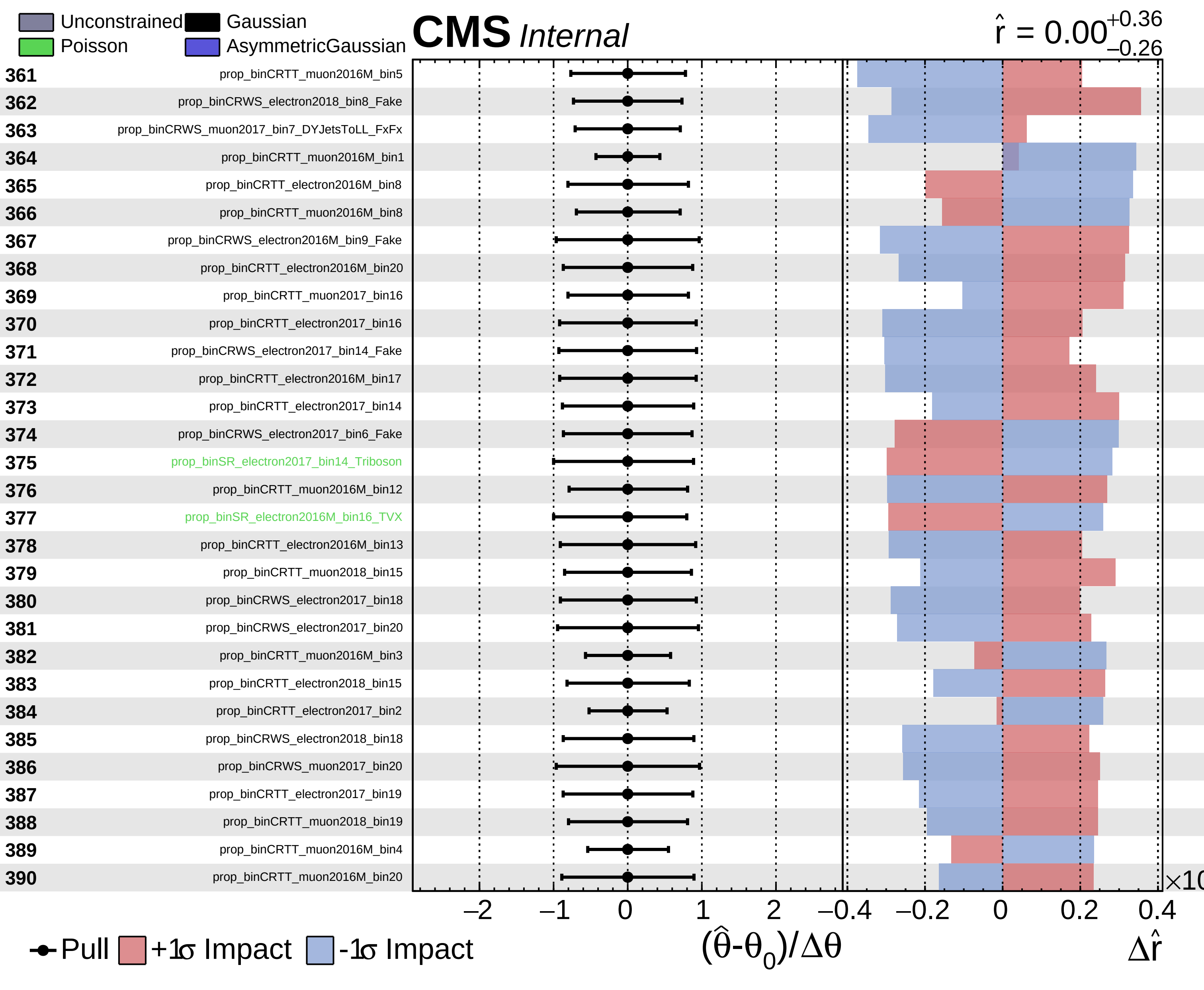
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.36}_{-0.26}$
271 prop_binSR_muon2018_bin0
272 prop_binCRTT_electron2018_bin8
273 prop_binSR_electron2016M_bin2_DYJetsToLL_FxFx
274 prop_binCRWS_electron2017_bin12
275 prop_binCRTT_muon2017_bin10
276 prop_binCRWS_electron2018_bin12_Fake
277 prop_binCRTT_muon2017_bin17
278 prop_binCRWS_electron2016M_bin5_Fake
279 prop_binCRTT_muon2018_bin9
280 prop_binCRTT_muon2017_bin20
281 prop_binCRTT_muon2018_bin14
282 prop_binSR_electron2018_bin0
283 ISR
284 prop_binCRTT_muon2017_bin1
285 prop_binCRWS_electron2017_bin5
286 prop_binCRWS_electron2018_bin4
287 prop_binCRWS_muon2016M_bin12
288 prop_binSR_electron2017_bin11_WrongSign
289 prop_binCRWS_muon2017_bin14_DYJetsToLL_FxFx
290 prop_binCRWS_electron2016M_bin12_DYJetsToLL_FxFx
291 prop_binCRTT_electron2018_bin14
292 prop_binCRWS_muon2017_bin11_Fake
293 prop_binCRTT_muon2018_bin18
294 prop_binCRTT_electron2016M_bin3
295 prop_binCRTT_electron2016M_bin6
296 prop_binCRTT_muon2018_bin1
297 prop_binCRTT_muon2018_bin5
298 prop_binCRTT_electron2018_bin12
299 prop_binCRTT_muon2018_bin7
300 prop_binCRTT_electron2016M_bin7
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

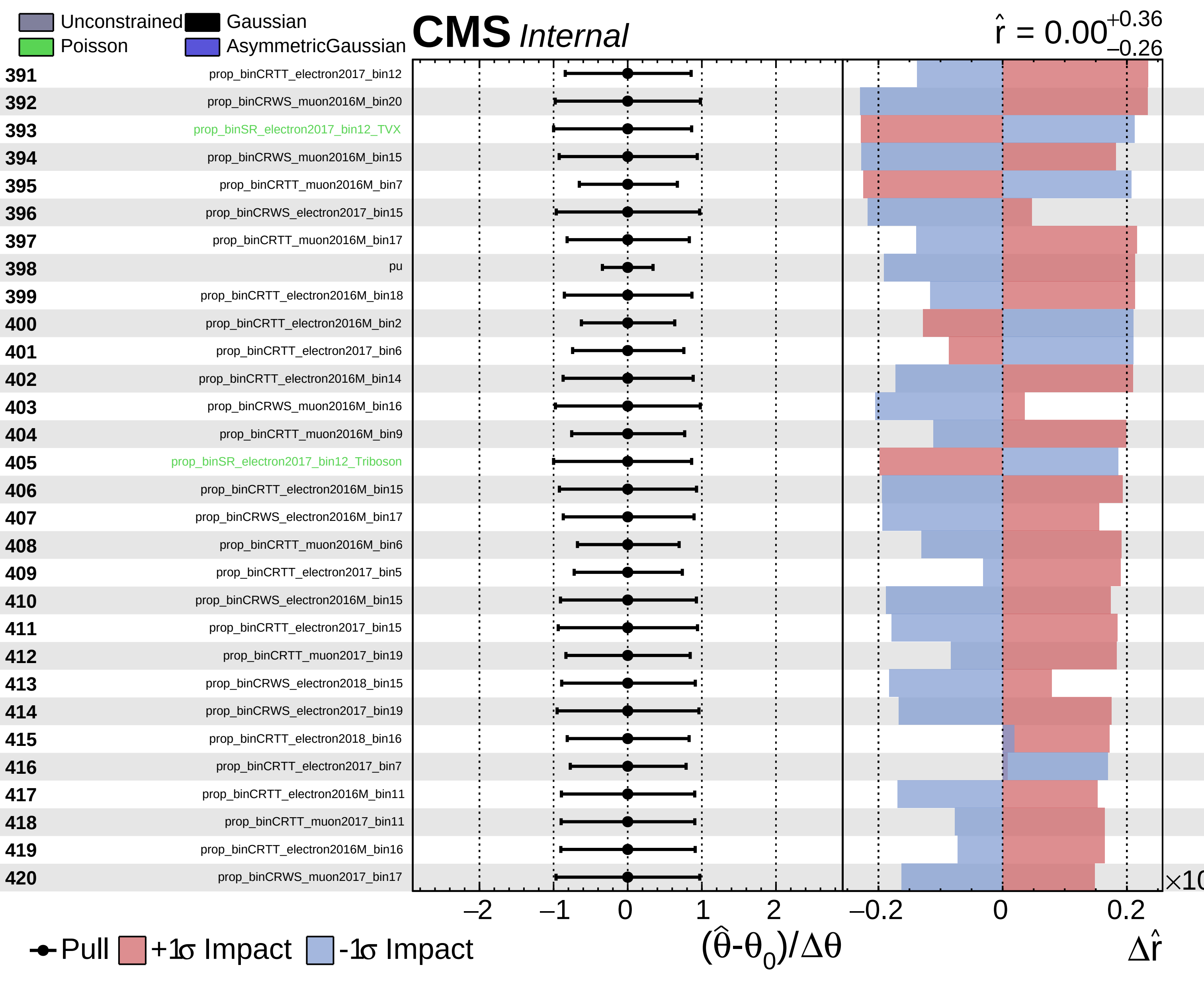


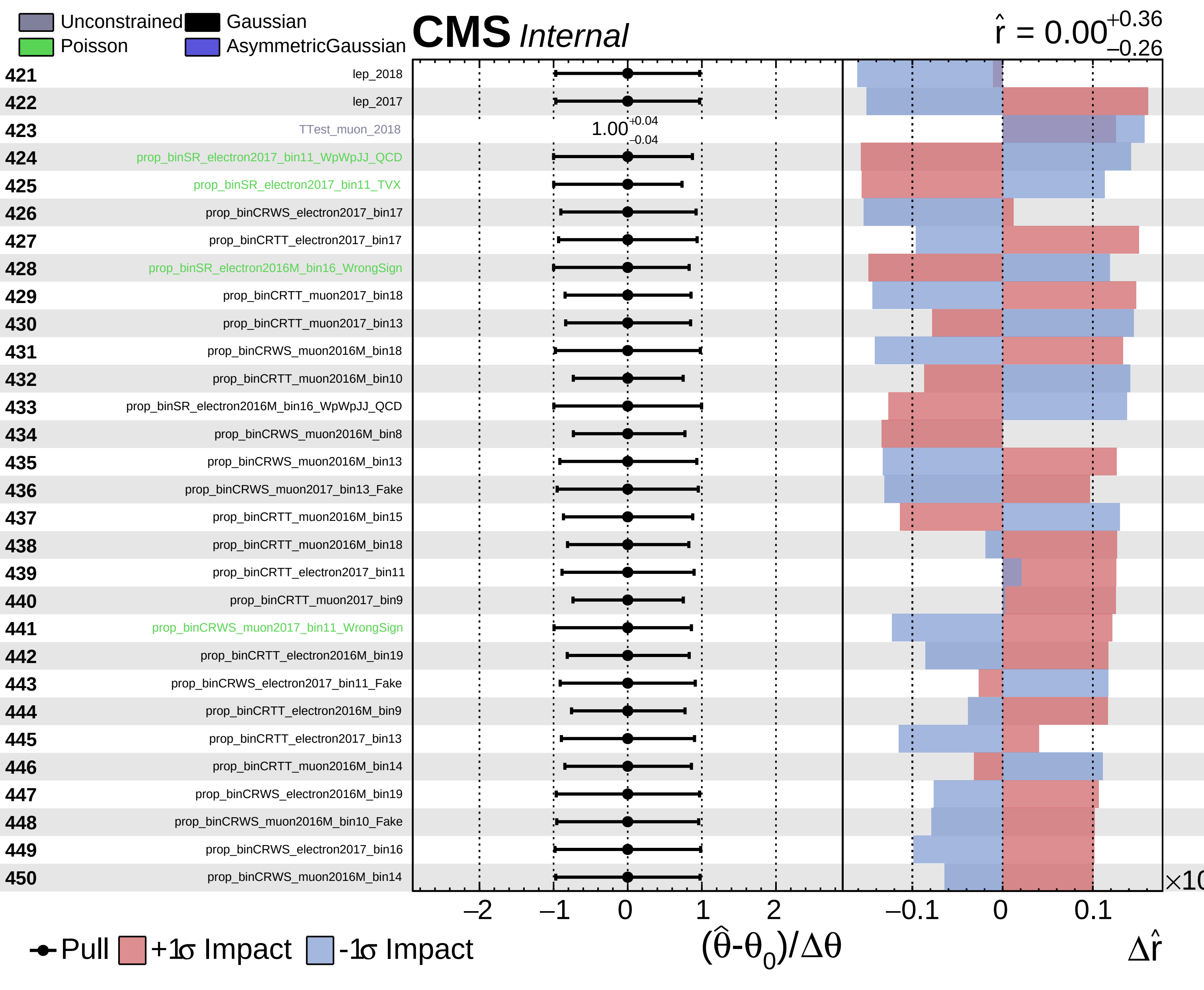
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.36}_{-0.26}$
301 prop_binCRWS_electron2016M_bin20
302 prop_binCRWS_electron2016M_bin7
303 prop_binCRWS_muon2018_bin16
304 prop_binCRTT_muon2016M_bin19
305 prop_binCRTT_muon2018_bin17
306 tau_vsjet_2018
307 prop_binCRWS_electron2017_bin11_DYJetsToLL_FxFx
308 prop_binCRTT_electron2018_bin18
309 prop_binCRWS_muon2017_bin5
310 prop_binCRWS_electron2018_bin6_DYJetsToLL_FxFx
311 prop_binSR_electron2017_bin14_WrongSign
312 prop_binCRTT_electron2017_bin20
313 tau_vsjet_2016M
314 prop_binCRTT_electron2017_bin18
315 TTest_muon_2017
$1.00^{+0.04}_{-0.04}$
316 prop_binCRWS_electron2017_bin14_DYJetsToLL_FxFx
317 prop_binSR_muon2017_bin13_WpWpJJ_QCD
318 prop_binSR_electron2016M_bin16_ZZtoLep
319 prop_binSR_electron2017_bin14_WpWpJJ_QCD
320 prop_binCRTT_electron2017_bin10
321 prop_binCRWS_muon2018_bin11
322 prop_binCRTT_muon2017_bin12
323 prop_binCRTT_electron2016M_bin10
324 prop_binCRWS_electron2018_bin6_Fake
325 TTest_electron_2017
$1.00^{+0.05}_{-0.05}$
326 prop_binCRWS_electron2016M_bin10_Fake
327 prop_binCRWS_electron2017_bin6_DYJetsToLL_FxFx
328 prop_binSR_electron2017_bin12_WpWpJJ_QCD
329 prop_binCRWS_electron2018_bin7
330 prop_binSR_electron2017_bin19_WrongSign
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



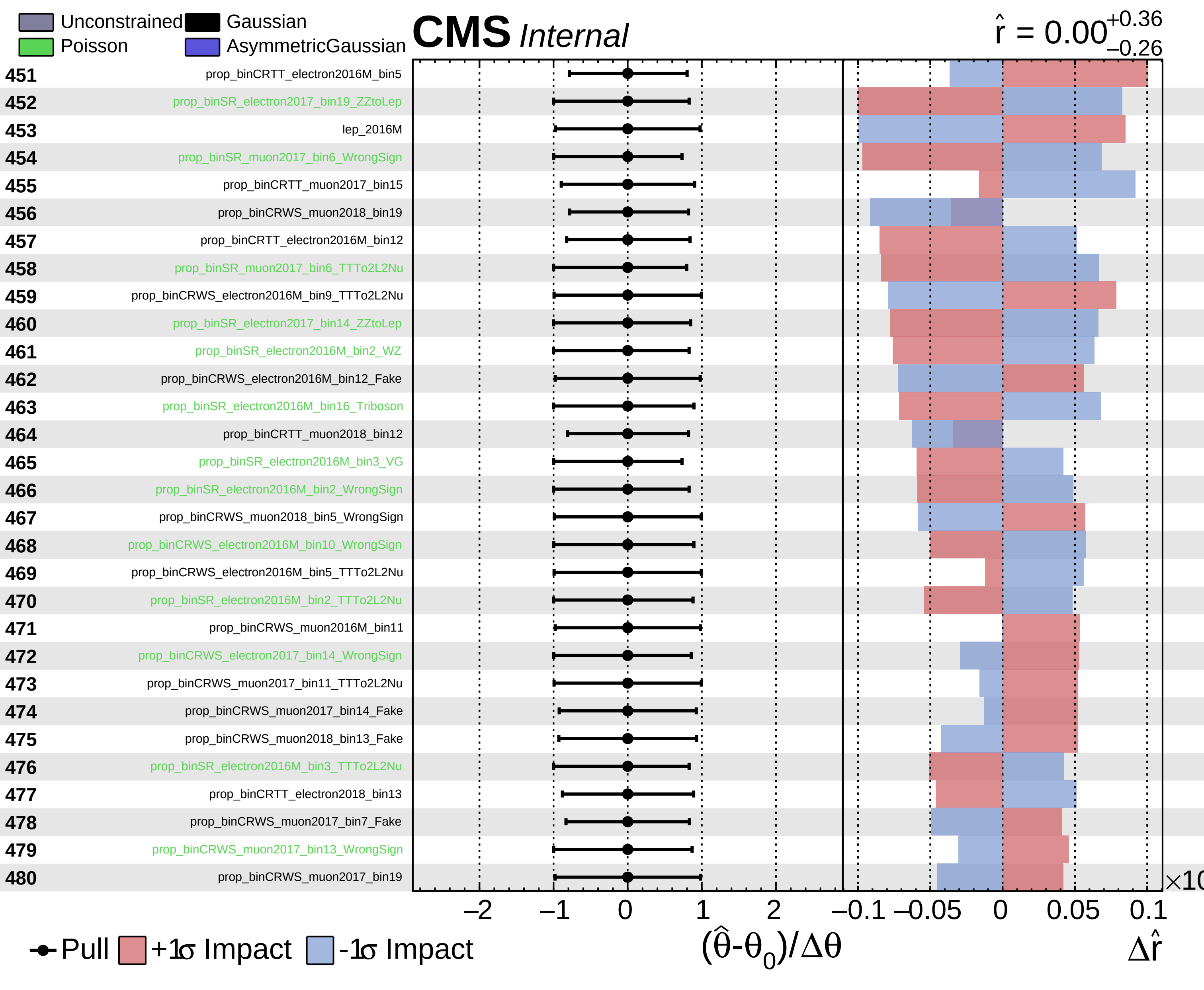
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.36}_{-0.26}$
331 prop_binCRTT_muon2017_bin8
332 prop_binCRTT_muon2017_bin4
333 prop_binSR_electron2017_bin0
334 tau_vsjet_2017
335 prop_binCRTT_muon2016M_bin2
336 prop_binCRTT_electron2018_bin17
337 prop_binCRTT_muon2017_bin14
338 prop_binSR_electron2017_bin11_TTTo2L2Nu
339 prop_binCRTT_muon2017_bin7
340 prop_binCRTT_muon2016M_bin11
341 prop_binSR_electron2017_bin12_WrongSign
342 prop_binCRTT_electron2018_bin7
343 prop_binCRTT_muon2016M_bin13
344 prop_binCRTT_muon2018_bin16
345 prop_binCRTT_electron2017_bin9
346 prop_binCRWS_muon2016M_bin19
347 TTest_electron_2018
$1.00^{+0.04}_{-0.04}$
348 prop_binCRTT_muon2017_bin2
349 prop_binCRWS_muon2017_bin16
350 prop_binCRTT_electron2018_bin6
351 prop_binCRTT_muon2018_bin8
352 prop_binSR_electron2016M_bin0
353 prop_binCRTT_electron2016M_bin4
354 prop_binSR_electron2017_bin19_Triboson
355 prop_binCRWS_muon2017_bin12_DYJetsToLL_FxFx
356 prop_binCRTT_electron2018_bin5
357 prop_binCRTT_electron2018_bin19
358 prop_binCRTT_electron2017_bin8
359 prop_binCRTT_electron2018_bin11
360 prop_binCRTT_muon2018_bin11
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.36}_{-0.26}$
361 prop_binCRTT_muon2016M_bin5
362 prop_binCRWS_electron2018_bin8_Fake
363 prop_binCRWS_muon2017_bin7_DYJetsToLL_FxFx
364 prop_binCRTT_muon2016M_bin1
365 prop_binCRTT_electron2016M_bin8
366 prop_binCRTT_muon2016M_bin8
367 prop_binCRWS_electron2016M_bin9_Fake
368 prop_binCRTT_electron2016M_bin20
369 prop_binCRTT_muon2017_bin16
370 prop_binCRTT_electron2017_bin16
371 prop_binCRWS_electron2017_bin14_Fake
372 prop_binCRTT_electron2016M_bin17
373 prop_binCRTT_electron2017_bin14
374 prop_binCRWS_electron2017_bin6_Fake
375 prop_binSR_electron2017_bin14_Triboson
376 prop_binCRTT_muon2016M_bin12
377 prop_binSR_electron2016M_bin16_TVX
378 prop_binCRTT_electron2016M_bin13
379 prop_binCRTT_muon2018_bin15
380 prop_binCRWS_electron2017_bin18
381 prop_binCRWS_electron2017_bin20
382 prop_binCRTT_muon2016M_bin3
383 prop_binCRTT_electron2018_bin15
384 prop_binCRTT_electron2017_bin2
385 prop_binCRWS_electron2018_bin18
386 prop_binCRWS_muon2017_bin20
387 prop_binCRTT_electron2017_bin19
388 prop_binCRTT_muon2018_bin19
389 prop_binCRTT_muon2016M_bin4
390 prop_binCRTT_muon2016M_bin20
−2 −1 0 1 2
−0.4 −0.2 0 0.2 0.4
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

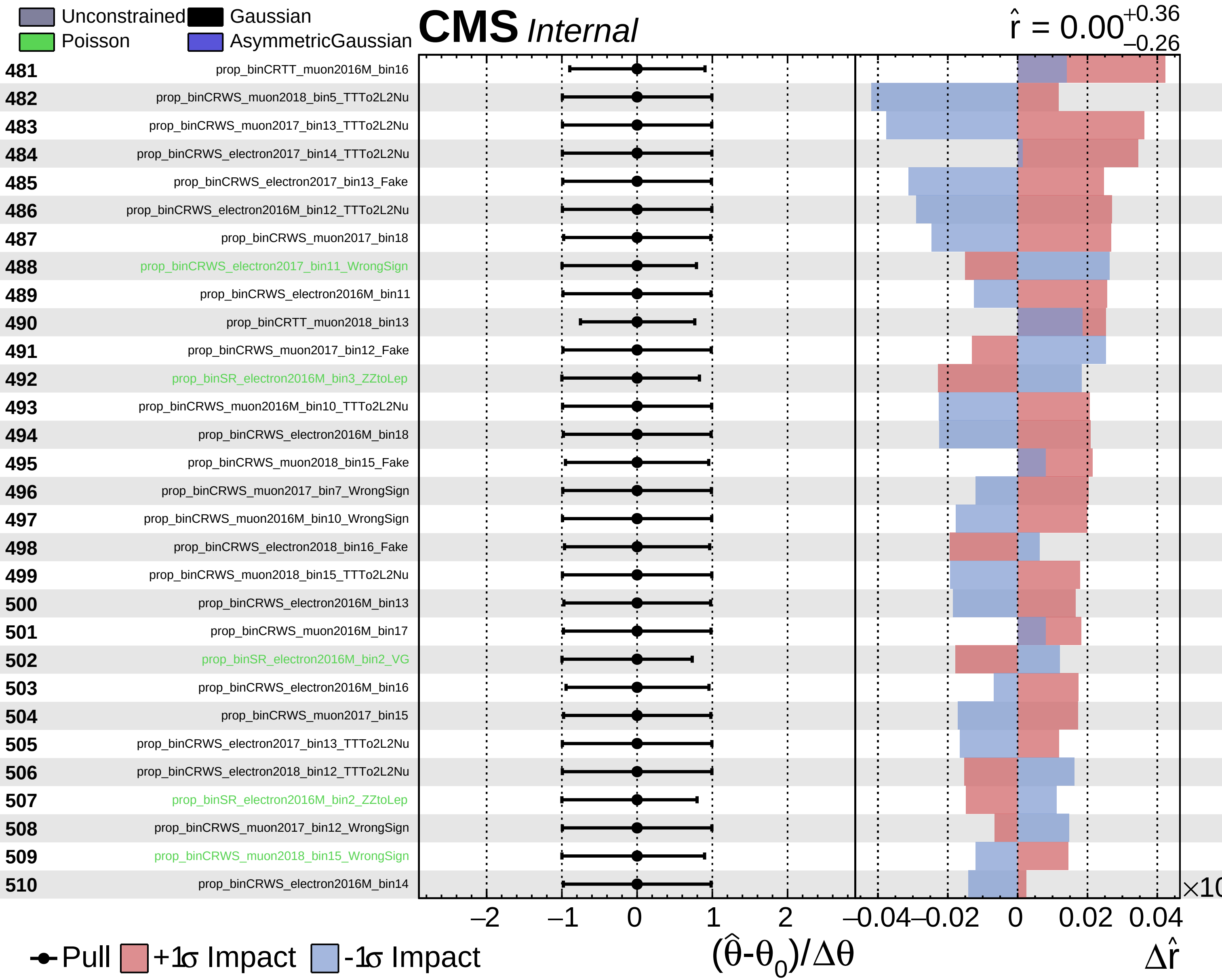

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.36}_{-0.26}$
391 prop_binCRTT_electron2017_bin12
392 prop_binCRWS_muon2016M_bin20
393 prop_binSR_electron2017_bin12_TVX
394 prop_binCRWS_muon2016M_bin15
395 prop_binCRTT_muon2016M_bin7
396 prop_binCRWS_electron2017_bin15
397 prop_binCRTT_muon2016M_bin17
398 pu
399 prop_binCRTT_electron2016M_bin18
400 prop_binCRTT_electron2016M_bin2
401 prop_binCRTT_electron2017_bin6
402 prop_binCRTT_electron2016M_bin14
403 prop_binCRWS_muon2016M_bin16
404 prop_binCRTT_muon2016M_bin9
405 prop_binSR_electron2017_bin12_Triboson
406 prop_binCRTT_electron2016M_bin15
407 prop_binCRWS_electron2016M_bin17
408 prop_binCRTT_muon2016M_bin6
409 prop_binCRTT_electron2017_bin5
410 prop_binCRWS_electron2016M_bin15
411 prop_binCRTT_electron2017_bin15
412 prop_binCRTT_muon2017_bin19
413 prop_binCRWS_electron2018_bin15
414 prop_binCRWS_electron2017_bin19
415 prop_binCRTT_electron2018_bin16
416 prop_binCRTT_electron2017_bin7
417 prop_binCRTT_electron2016M_bin11
418 prop_binCRTT_muon2017_bin11
419 prop_binCRTT_electron2016M_bin16
420 prop_binCRWS_muon2017_bin17
−2 −1 0 1 2
−0.2 0 0.2
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

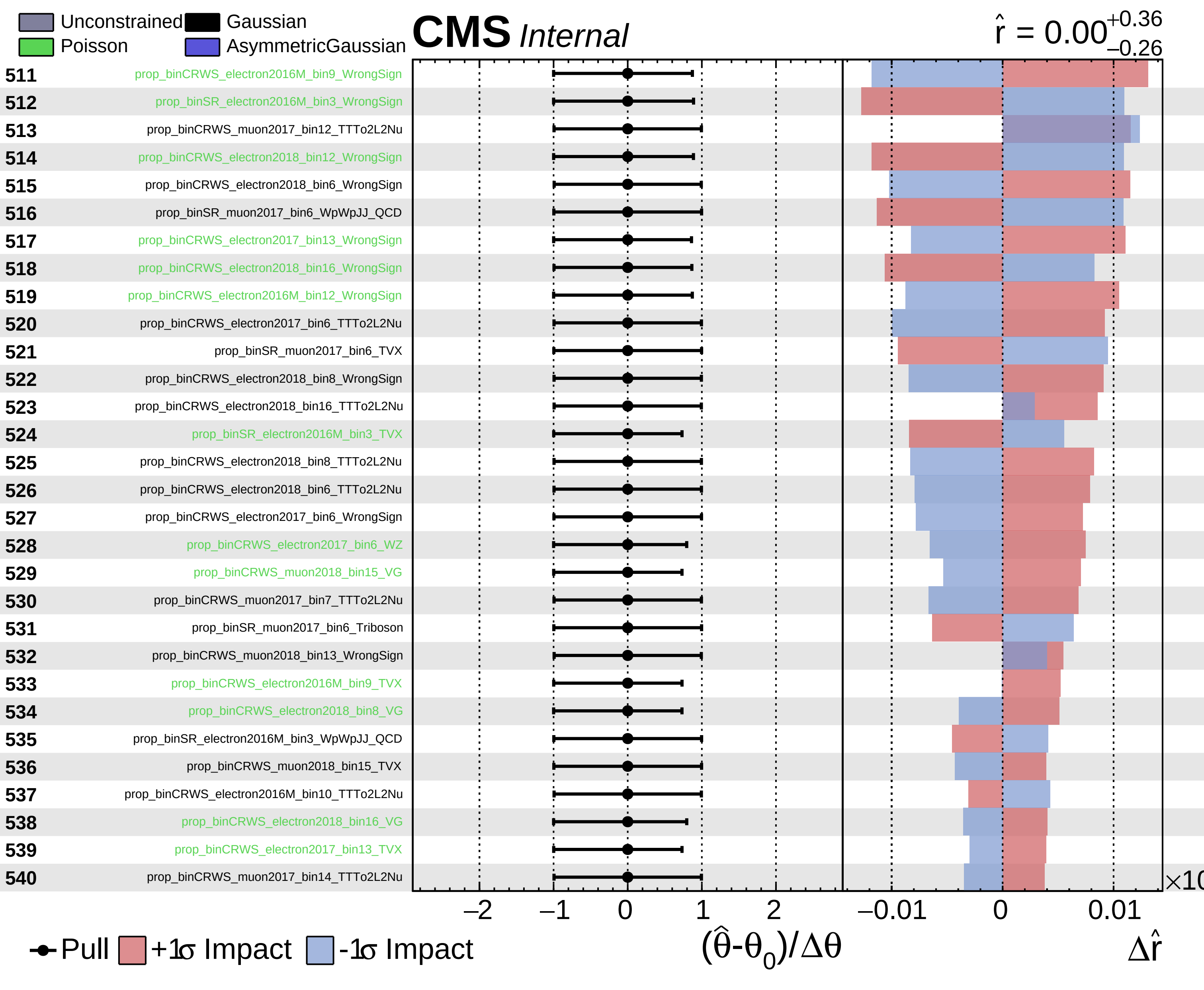


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.36 −0.26
421 lep_2018
422 lep_2017
423 TTest_muon_2018
1.00 +0.04 −0.04
424 prop_binSR_electron2017_bin11_WpWpJJ_QCD
425 prop_binSR_electron2017_bin11_TVX
426 prop_binCRWS_electron2017_bin17
427 prop_binCRTT_electron2017_bin17
428 prop_binSR_electron2016M_bin16_WrongSign
429 prop_binCRTT_muon2017_bin18
430 prop_binCRTT_muon2017_bin13
431 prop_binCRWS_muon2016M_bin18
432 prop_binCRTT_muon2016M_bin10
433 prop_binSR_electron2016M_bin16_WpWpJJ_QCD
434 prop_binCRWS_muon2016M_bin8
435 prop_binCRWS_muon2016M_bin13
436 prop_binCRWS_muon2017_bin13_Fake
437 prop_binCRTT_muon2016M_bin15
438 prop_binCRTT_muon2016M_bin18
439 prop_binCRTT_electron2017_bin11
440 prop_binCRTT_muon2017_bin9
441 prop_binCRWS_muon2017_bin11_WrongSign
442 prop_binCRTT_electron2016M_bin19
443 prop_binCRWS_electron2017_bin11_Fake
444 prop_binCRTT_electron2016M_bin9
445 prop_binCRTT_electron2017_bin13
446 prop_binCRTT_muon2016M_bin14
447 prop_binCRWS_electron2016M_bin19
448 prop_binCRWS_muon2016M_bin10_Fake
449 prop_binCRWS_electron2017_bin16
450 prop_binCRWS_muon2016M_bin14
−2 −1 0 1 2
−0.1 0 0.1
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂

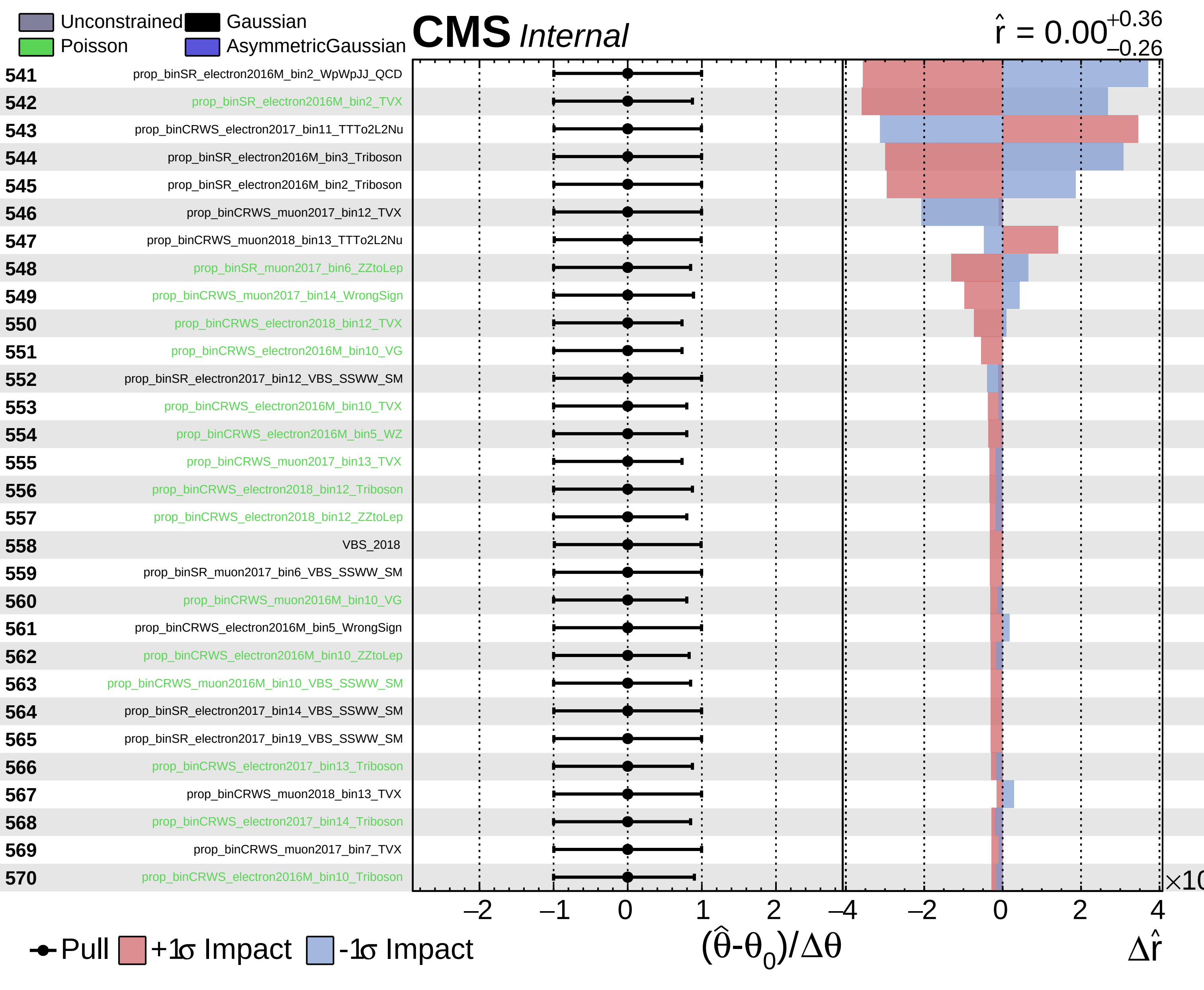


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.36 −0.26
451 prop_binCRTT_electron2016M_bin5
452 prop_binSR_electron2017_bin19_ZZtoLep
453 lep_2016M
454 prop_binSR_muon2017_bin6_WrongSign
455 prop_binCRTT_muon2017_bin15
456 prop_binCRWS_muon2018_bin19
457 prop_binCRTT_electron2016M_bin12
458 prop_binSR_muon2017_bin6_TTTo2L2Nu
459 prop_binCRWS_electron2016M_bin9_TTTo2L2Nu
460 prop_binSR_electron2017_bin14_ZZtoLep
461 prop_binSR_electron2016M_bin2_WZ
462 prop_binCRWS_electron2016M_bin12_Fake
463 prop_binSR_electron2016M_bin16_Triboson
464 prop_binCRTT_muon2018_bin12
465 prop_binSR_electron2016M_bin3_VG
466 prop_binSR_electron2016M_bin2_WrongSign
467 prop_binCRWS_muon2018_bin5_WrongSign
468 prop_binCRWS_electron2016M_bin10_WrongSign
469 prop_binCRWS_electron2016M_bin5_TTTo2L2Nu
470 prop_binSR_electron2016M_bin2_TTTo2L2Nu
471 prop_binCRWS_muon2016M_bin11
472 prop_binCRWS_electron2017_bin14_WrongSign
473 prop_binCRWS_muon2017_bin11_TTTo2L2Nu
474 prop_binCRWS_muon2017_bin14_Fake
475 prop_binCRWS_muon2018_bin13_Fake
476 prop_binSR_electron2016M_bin3_TTTo2L2Nu
477 prop_binCRTT_electron2018_bin13
478 prop_binCRWS_muon2017_bin7_Fake
479 prop_binCRWS_muon2017_bin13_WrongSign
480 prop_binCRWS_muon2017_bin19
−2 −1 0 1 2
−0.1 −0.05 0 0.05 0.1
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂

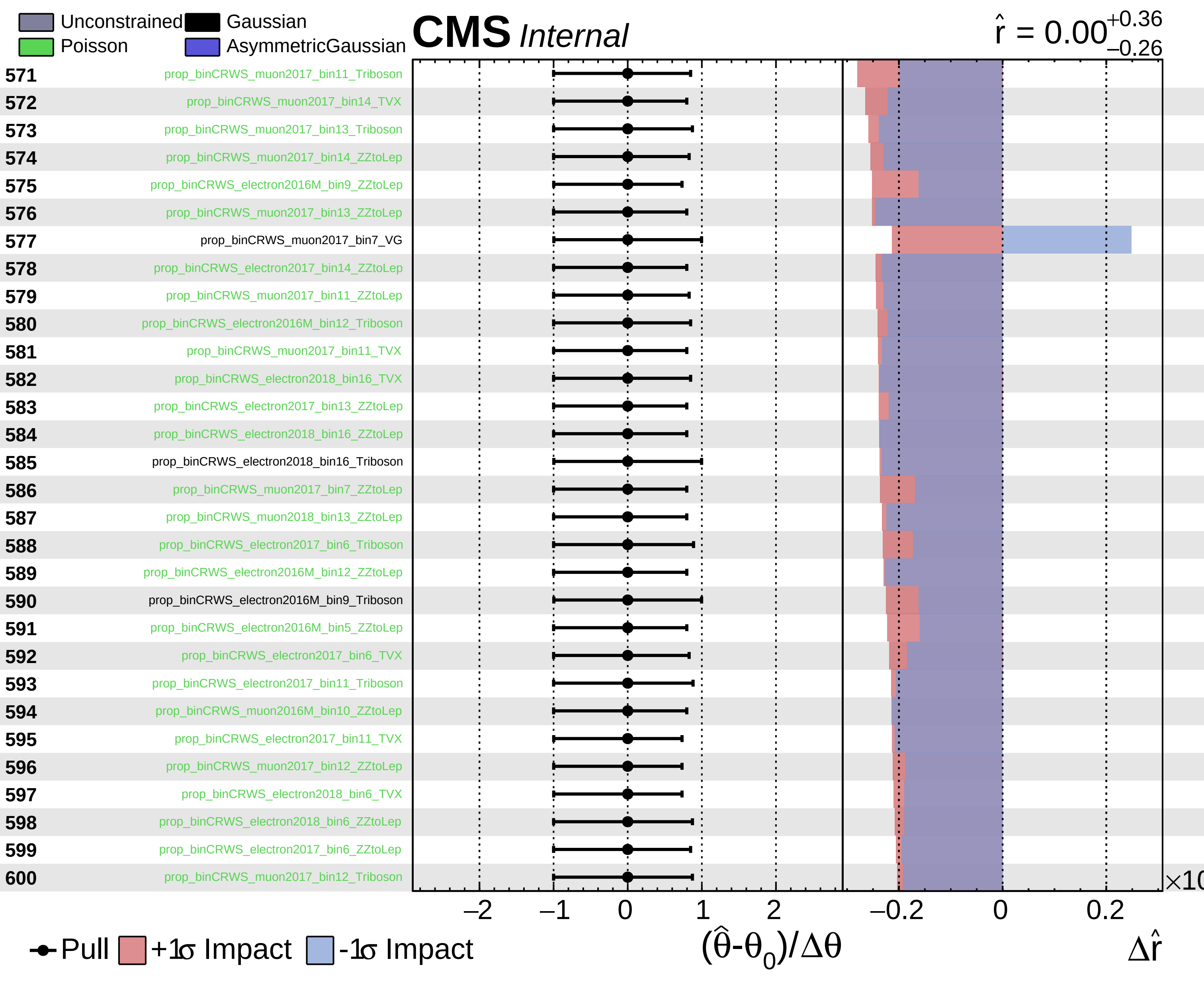

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.36 −0.26
481 prop_binCRTT_muon2016M_bin16
482 prop_binCRWS_muon2018_bin5_TTTo2L2Nu
483 prop_binCRWS_muon2017_bin13_TTTo2L2Nu
484 prop_binCRWS_electron2017_bin14_TTTo2L2Nu
485 prop_binCRWS_electron2017_bin13_Fake
486 prop_binCRWS_electron2016M_bin12_TTTo2L2Nu
487 prop_binCRWS_muon2017_bin18
488 prop_binCRWS_electron2017_bin11_WrongSign
489 prop_binCRWS_electron2016M_bin11
490 prop_binCRTT_muon2018_bin13
491 prop_binCRWS_muon2017_bin12_Fake
492 prop_binSR_electron2016M_bin3_ZZtoLep
493 prop_binCRWS_muon2016M_bin10_TTTo2L2Nu
494 prop_binCRWS_electron2016M_bin18
495 prop_binCRWS_muon2018_bin15_Fake
496 prop_binCRWS_muon2017_bin7_WrongSign
497 prop_binCRWS_muon2016M_bin10_WrongSign
498 prop_binCRWS_electron2018_bin16_Fake
499 prop_binCRWS_muon2018_bin15_TTTo2L2Nu
500 prop_binCRWS_electron2016M_bin13
501 prop_binCRWS_muon2016M_bin17
502 prop_binSR_electron2016M_bin2_VG
503 prop_binCRWS_electron2016M_bin16
504 prop_binCRWS_muon2017_bin15
505 prop_binCRWS_electron2017_bin13_TTTo2L2Nu
506 prop_binCRWS_electron2018_bin12_TTTo2L2Nu
507 prop_binSR_electron2016M_bin2_ZZtoLep
508 prop_binCRWS_muon2017_bin12_WrongSign
509 prop_binCRWS_muon2018_bin15_WrongSign
510 prop_binCRWS_electron2016M_bin14
−2 −1 0 1 2
−0.04 −0.02 0 0.02 0.04
×10
Pull +1σ Impact -1σ Impact
(θ̂-θ0)/Δθ
Δr̂

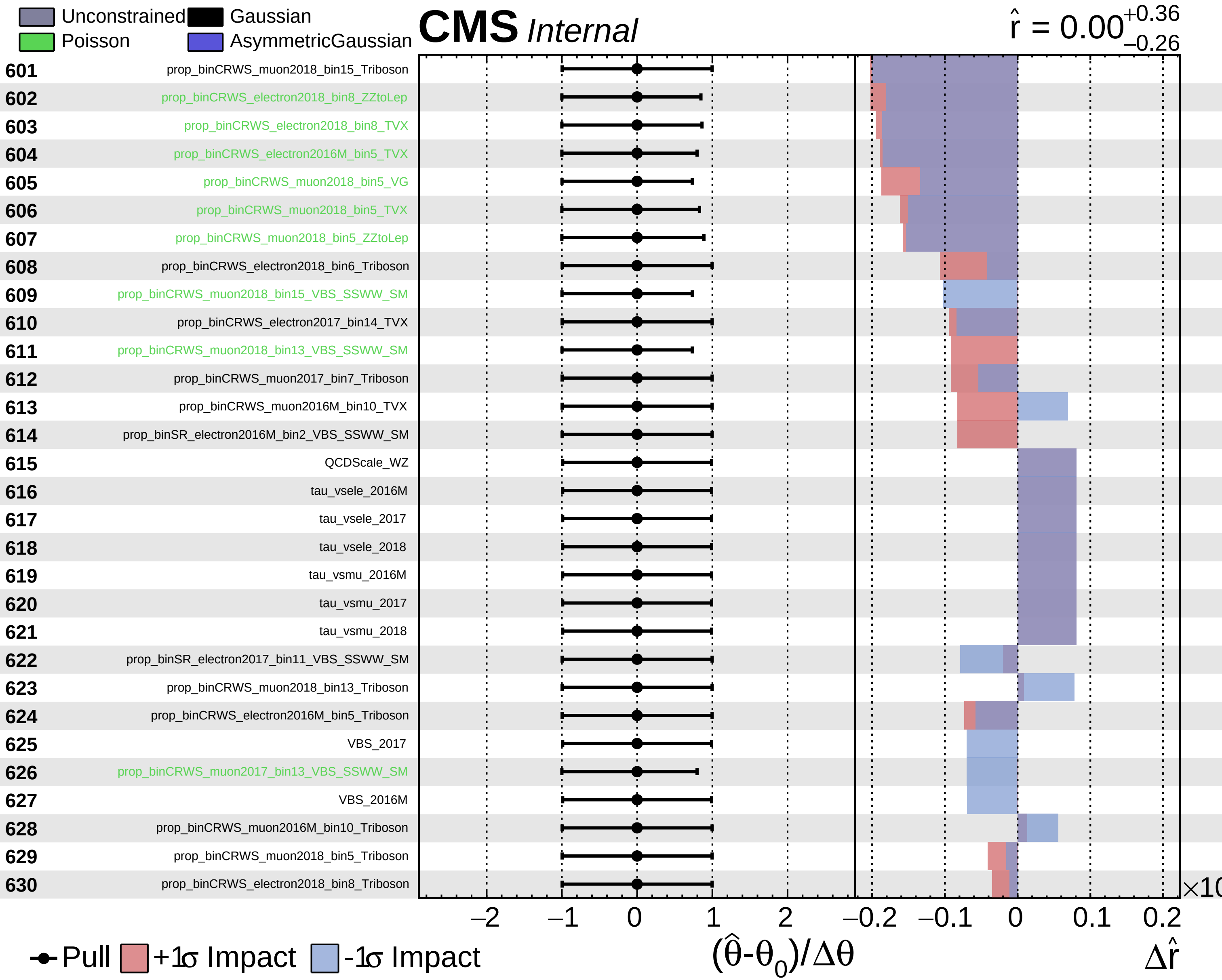


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.36}_{-0.26}$
511 prop_binCRWS_electron2016M_bin9_WrongSign
512 prop_binSR_electron2016M_bin3_WrongSign
513 prop_binCRWS_muon2017_bin12_TTTo2L2Nu
514 prop_binCRWS_electron2018_bin12_WrongSign
515 prop_binCRWS_electron2018_bin6_WrongSign
516 prop_binSR_muon2017_bin6_WpWpJJ_QCD
517 prop_binCRWS_electron2017_bin13_WrongSign
518 prop_binCRWS_electron2018_bin16_WrongSign
519 prop_binCRWS_electron2016M_bin12_WrongSign
520 prop_binCRWS_electron2017_bin6_TTTo2L2Nu
521 prop_binSR_muon2017_bin6_TVX
522 prop_binCRWS_electron2018_bin8_WrongSign
523 prop_binCRWS_electron2018_bin16_TTTo2L2Nu
524 prop_binSR_electron2016M_bin3_TVX
525 prop_binCRWS_electron2018_bin8_TTTo2L2Nu
526 prop_binCRWS_electron2018_bin6_TTTo2L2Nu
527 prop_binCRWS_electron2017_bin6_WrongSign
528 prop_binCRWS_electron2017_bin6_WZ
529 prop_binCRWS_muon2018_bin15_VG
530 prop_binCRWS_muon2017_bin7_TTTo2L2Nu
531 prop_binSR_muon2017_bin6_Triboson
532 prop_binCRWS_muon2018_bin13_WrongSign
533 prop_binCRWS_electron2016M_bin9_TVX
534 prop_binCRWS_electron2018_bin8_VG
535 prop_binSR_electron2016M_bin3_WpWpJJ_QCD
536 prop_binCRWS_muon2018_bin15_TVX
537 prop_binCRWS_electron2016M_bin10_TTTo2L2Nu
538 prop_binCRWS_electron2018_bin16_VG
539 prop_binCRWS_electron2017_bin13_TVX
540 prop_binCRWS_muon2017_bin14_TTTo2L2Nu
−2
−1
0
1
2
−0.01
0
0.01
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

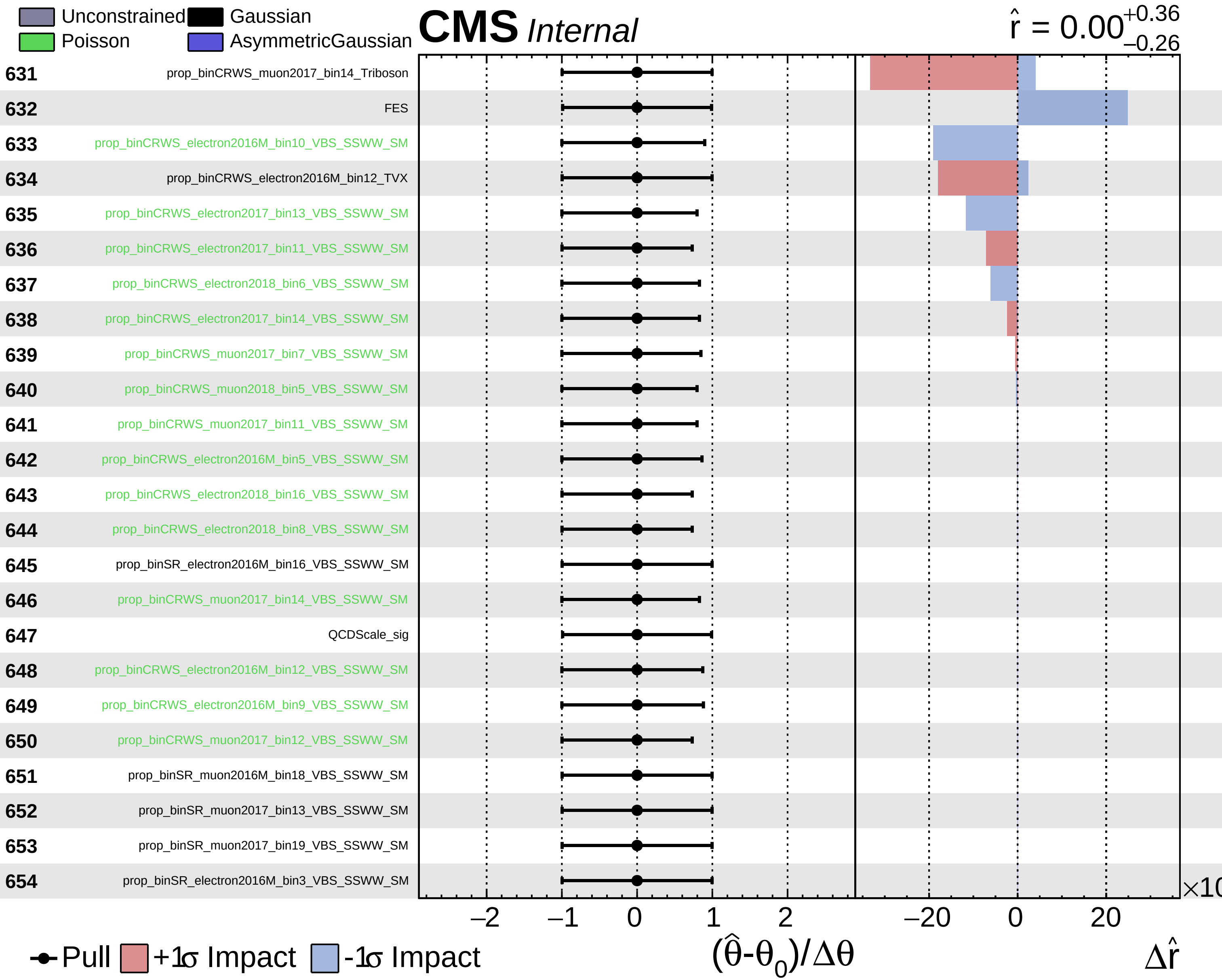


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.36}_{-0.26}$
541 prop_binSR_electron2016M_bin2_WpWpJJ_QCD
542 prop_binSR_electron2016M_bin2_TVX
543 prop_binCRWS_electron2017_bin11_TTTo2L2Nu
544 prop_binSR_electron2016M_bin3_Triboson
545 prop_binSR_electron2016M_bin2_Triboson
546 prop_binCRWS_muon2017_bin12_TVX
547 prop_binCRWS_muon2018_bin13_TTTo2L2Nu
548 prop_binSR_muon2017_bin6_ZZtoLep
549 prop_binCRWS_muon2017_bin14_WrongSign
550 prop_binCRWS_electron2018_bin12_TVX
551 prop_binCRWS_electron2016M_bin10_VG
552 prop_binSR_electron2017_bin12_VBS_SSWW_SM
553 prop_binCRWS_electron2016M_bin10_TVX
554 prop_binCRWS_electron2016M_bin5_WZ
555 prop_binCRWS_muon2017_bin13_TVX
556 prop_binCRWS_electron2018_bin12_Triboson
557 prop_binCRWS_electron2018_bin12_ZZtoLep
558 VBS_2018
559 prop_binSR_muon2017_bin6_VBS_SSWW_SM
560 prop_binCRWS_muon2016M_bin10_VG
561 prop_binCRWS_electron2016M_bin5_WrongSign
562 prop_binCRWS_electron2016M_bin10_ZZtoLep
563 prop_binCRWS_muon2016M_bin10_VBS_SSWW_SM
564 prop_binSR_electron2017_bin14_VBS_SSWW_SM
565 prop_binSR_electron2017_bin19_VBS_SSWW_SM
566 prop_binCRWS_electron2017_bin13_Triboson
567 prop_binCRWS_muon2018_bin13_TVX
568 prop_binCRWS_electron2017_bin14_Triboson
569 prop_binCRWS_muon2017_bin7_TVX
570 prop_binCRWS_electron2016M_bin10_Triboson
−2 −1 0 1 2
−4 −2 0 2 4
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.36 −0.26
571 prop_binCRWS_muon2017_bin11_Triboson
572 prop_binCRWS_muon2017_bin14_TVX
573 prop_binCRWS_muon2017_bin13_Triboson
574 prop_binCRWS_muon2017_bin14_ZZtoLep
575 prop_binCRWS_electron2016M_bin9_ZZtoLep
576 prop_binCRWS_muon2017_bin13_ZZtoLep
577 prop_binCRWS_muon2017_bin7_VG
578 prop_binCRWS_electron2017_bin14_ZZtoLep
579 prop_binCRWS_muon2017_bin11_ZZtoLep
580 prop_binCRWS_electron2016M_bin12_Triboson
581 prop_binCRWS_muon2017_bin11_TVX
582 prop_binCRWS_electron2018_bin16_TVX
583 prop_binCRWS_electron2017_bin13_ZZtoLep
584 prop_binCRWS_electron2018_bin16_ZZtoLep
585 prop_binCRWS_electron2018_bin16_Triboson
586 prop_binCRWS_muon2017_bin7_ZZtoLep
587 prop_binCRWS_muon2018_bin13_ZZtoLep
588 prop_binCRWS_electron2017_bin6_Triboson
589 prop_binCRWS_electron2016M_bin12_ZZtoLep
590 prop_binCRWS_electron2016M_bin9_Triboson
591 prop_binCRWS_electron2016M_bin5_ZZtoLep
592 prop_binCRWS_electron2017_bin6_TVX
593 prop_binCRWS_electron2017_bin11_Triboson
594 prop_binCRWS_muon2016M_bin10_ZZtoLep
595 prop_binCRWS_electron2017_bin11_TVX
596 prop_binCRWS_muon2017_bin12_ZZtoLep
597 prop_binCRWS_electron2018_bin6_TVX
598 prop_binCRWS_electron2018_bin6_ZZtoLep
599 prop_binCRWS_electron2017_bin6_ZZtoLep
600 prop_binCRWS_muon2017_bin12_Triboson
−2
−1
0
1
2
−0.2
0
0.2
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.36 −0.26
601 prop_binCRWS_muon2018_bin15_Triboson
602 prop_binCRWS_electron2018_bin8_ZZtoLep
603 prop_binCRWS_electron2018_bin8_TVX
604 prop_binCRWS_electron2016M_bin5_TVX
605 prop_binCRWS_muon2018_bin5_VG
606 prop_binCRWS_muon2018_bin5_TVX
607 prop_binCRWS_muon2018_bin5_ZZtoLep
608 prop_binCRWS_electron2018_bin6_Triboson
609 prop_binCRWS_muon2018_bin15_VBS_SSWW_SM
610 prop_binCRWS_electron2017_bin14_TVX
611 prop_binCRWS_muon2018_bin13_VBS_SSWW_SM
612 prop_binCRWS_muon2017_bin7_Triboson
613 prop_binCRWS_muon2016M_bin10_TVX
614 prop_binSR_electron2016M_bin2_VBS_SSWW_SM
615 QCDScale_WZ
616 tau_vsele_2016M
617 tau_vsele_2017
618 tau_vsele_2018
619 tau_vsmu_2016M
620 tau_vsmu_2017
621 tau_vsmu_2018
622 prop_binSR_electron2017_bin11_VBS_SSWW_SM
623 prop_binCRWS_muon2018_bin13_Triboson
624 prop_binCRWS_electron2016M_bin5_Triboson
625 VBS_2017
626 prop_binCRWS_muon2017_bin13_VBS_SSWW_SM
627 VBS_2016M
628 prop_binCRWS_muon2016M_bin10_Triboson
629 prop_binCRWS_muon2018_bin5_Triboson
630 prop_binCRWS_electron2018_bin8_Triboson
−2 −1 0 1 2
−0.2 −0.1 0 0.1 0.2
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.36 −0.26
631 prop_binCRWS_muon2017_bin14_Triboson
632 FES
633 prop_binCRWS_electron2016M_bin10_VBS_SSWW_SM
634 prop_binCRWS_electron2016M_bin12_TVX
635 prop_binCRWS_electron2017_bin13_VBS_SSWW_SM
636 prop_binCRWS_electron2017_bin11_VBS_SSWW_SM
637 prop_binCRWS_electron2018_bin6_VBS_SSWW_SM
638 prop_binCRWS_electron2017_bin14_VBS_SSWW_SM
639 prop_binCRWS_muon2017_bin7_VBS_SSWW_SM
640 prop_binCRWS_muon2018_bin5_VBS_SSWW_SM
641 prop_binCRWS_muon2017_bin11_VBS_SSWW_SM
642 prop_binCRWS_electron2016M_bin5_VBS_SSWW_SM
643 prop_binCRWS_electron2018_bin16_VBS_SSWW_SM
644 prop_binCRWS_electron2018_bin8_VBS_SSWW_SM
645 prop_binSR_electron2016M_bin16_VBS_SSWW_SM
646 prop_binCRWS_muon2017_bin14_VBS_SSWW_SM
647 QCDScale_sig
648 prop_binCRWS_electron2016M_bin12_VBS_SSWW_SM
649 prop_binCRWS_electron2016M_bin9_VBS_SSWW_SM
650 prop_binCRWS_muon2017_bin12_VBS_SSWW_SM
651 prop_binSR_muon2016M_bin18_VBS_SSWW_SM
652 prop_binSR_muon2017_bin13_VBS_SSWW_SM
653 prop_binSR_muon2017_bin19_VBS_SSWW_SM
654 prop_binSR_electron2016M_bin3_VBS_SSWW_SM
−2 −1 0 1 2
−20 0 20
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂